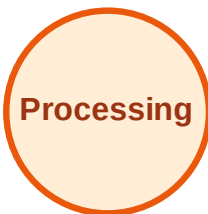


BFS Traversal

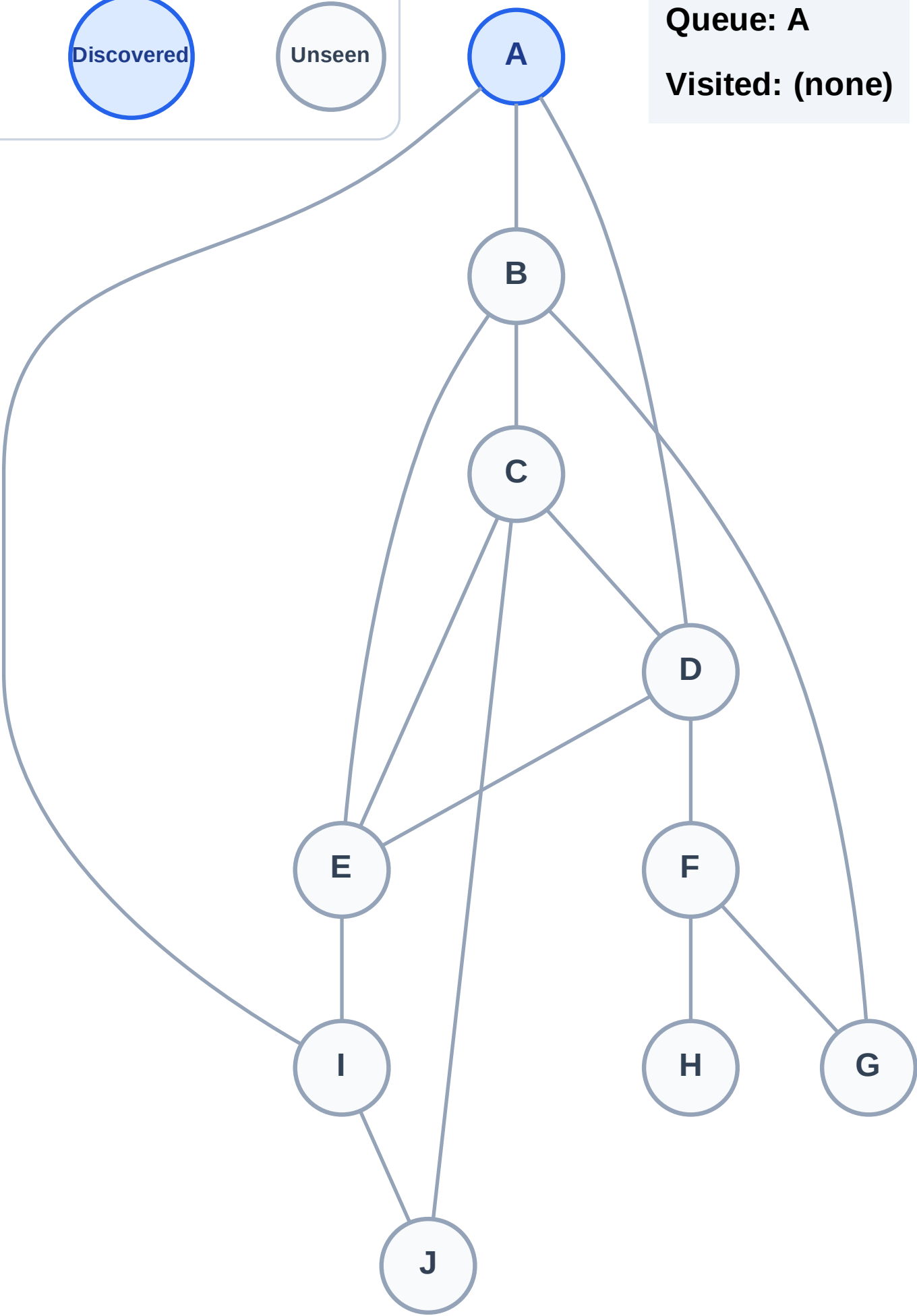
Step 1: Start at A

Legend



Queue: A

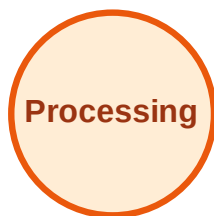
Visited: (none)



BFS Traversal

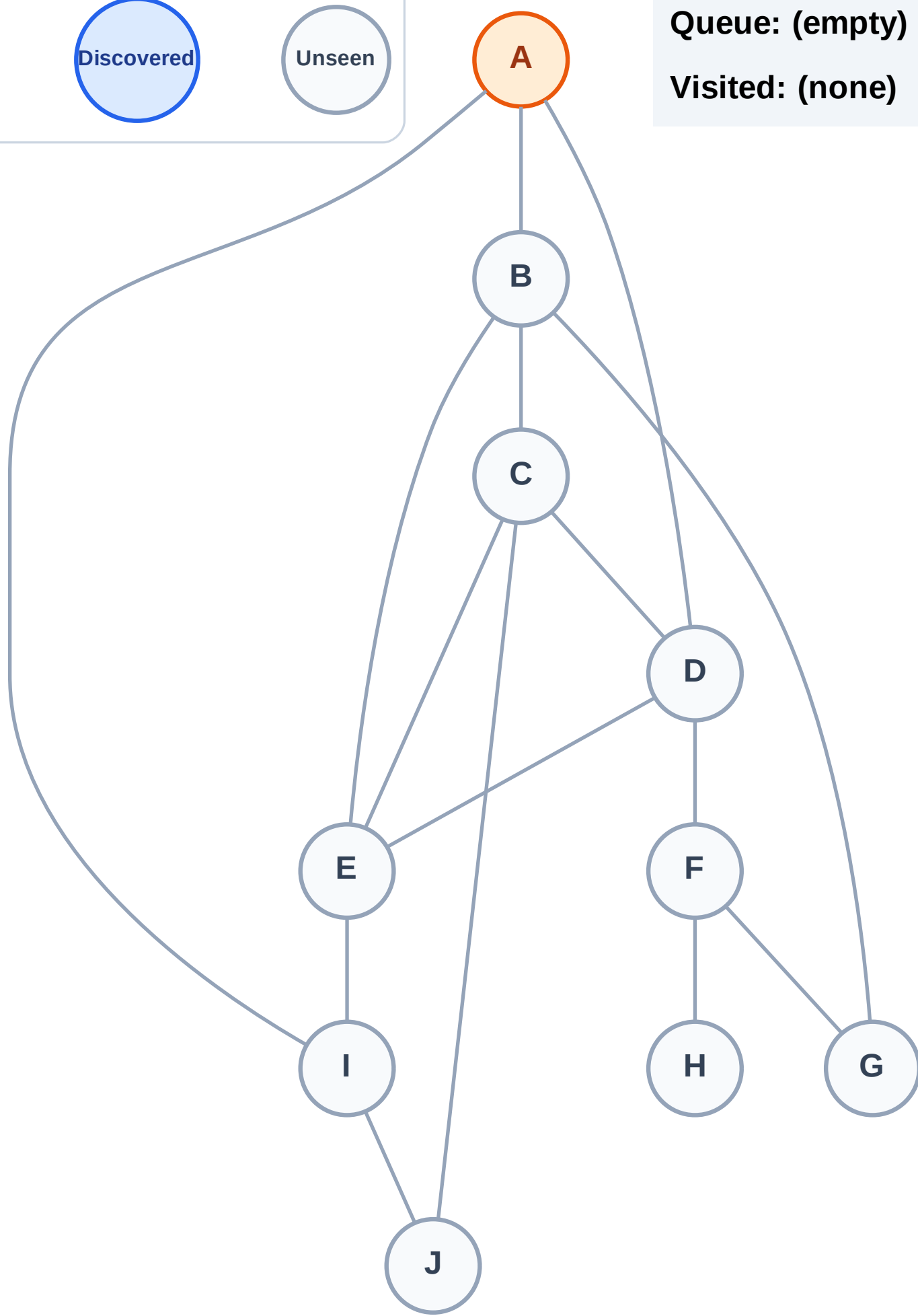
Step 2: Process node A

Legend



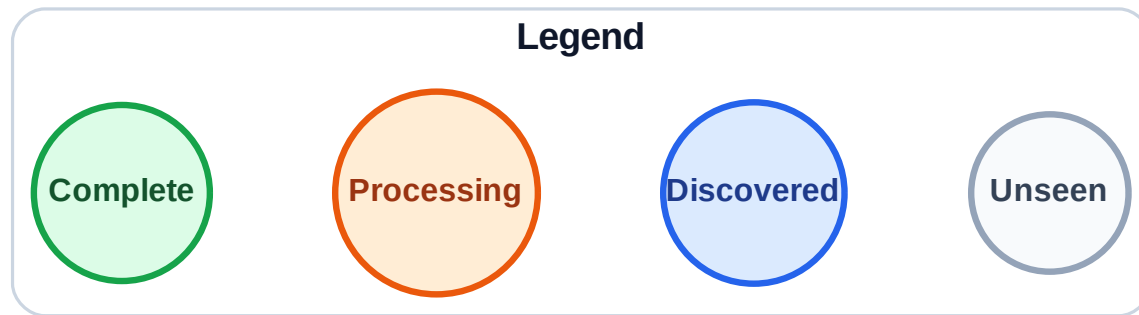
Queue: (empty)

Visited: (none)

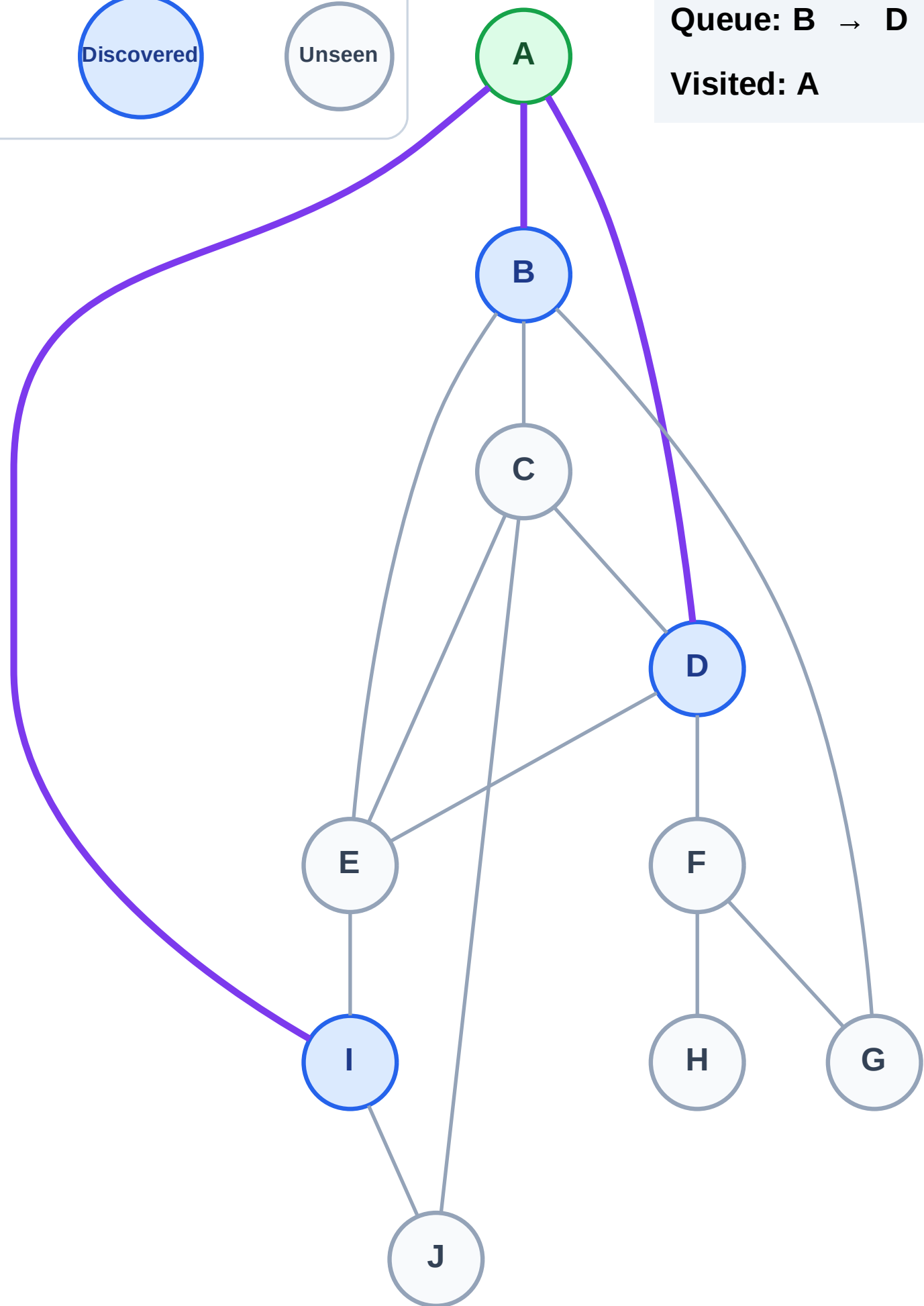


BFS Traversal

Step 3: Discover B, D, I from A



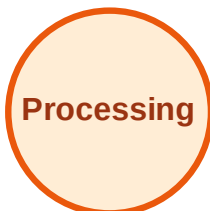
Queue: B → D → I
Visited: A



BFS Traversal

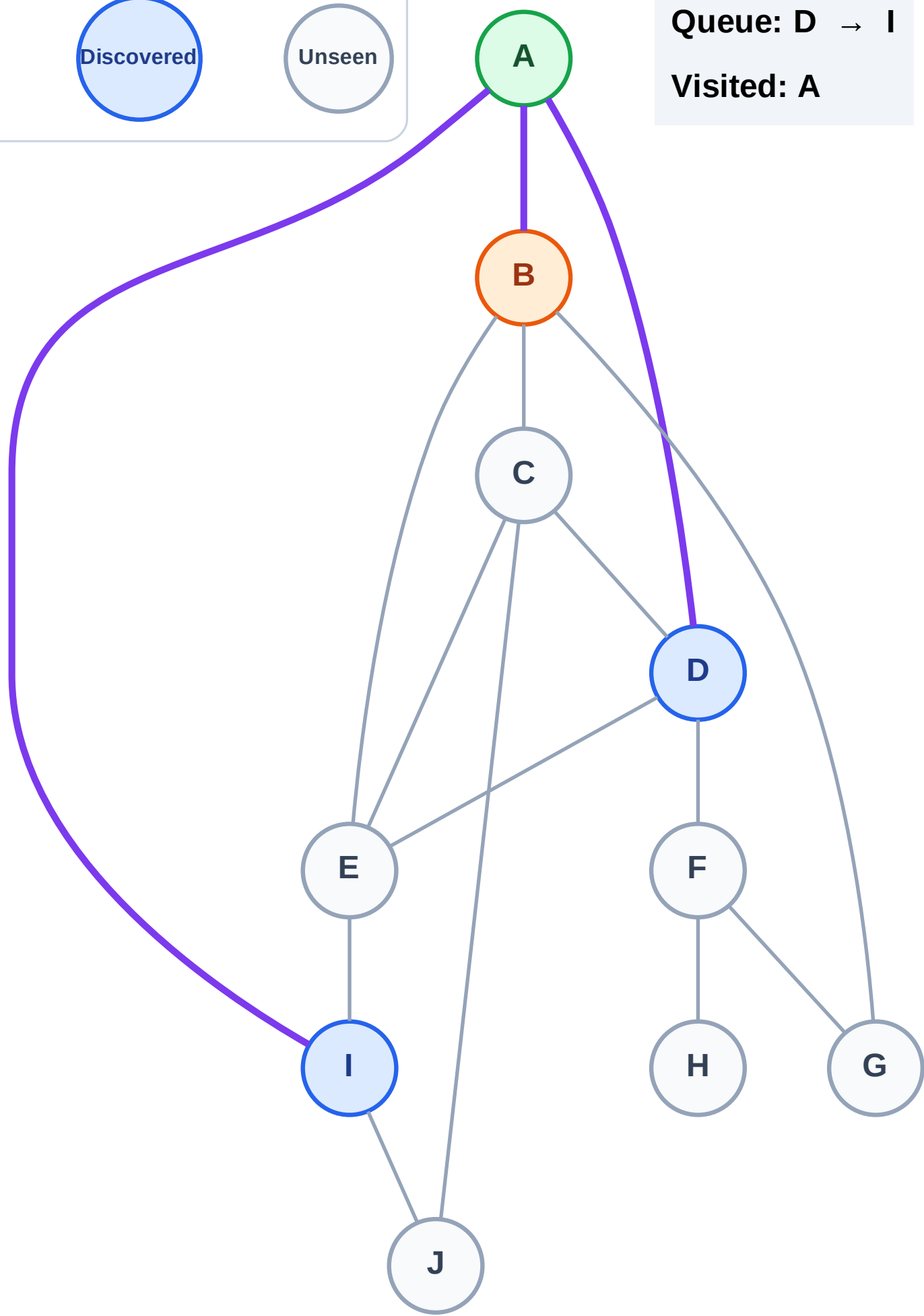
Step 4: Process node B

Legend



Queue: D → I

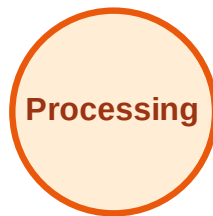
Visited: A



BFS Traversal

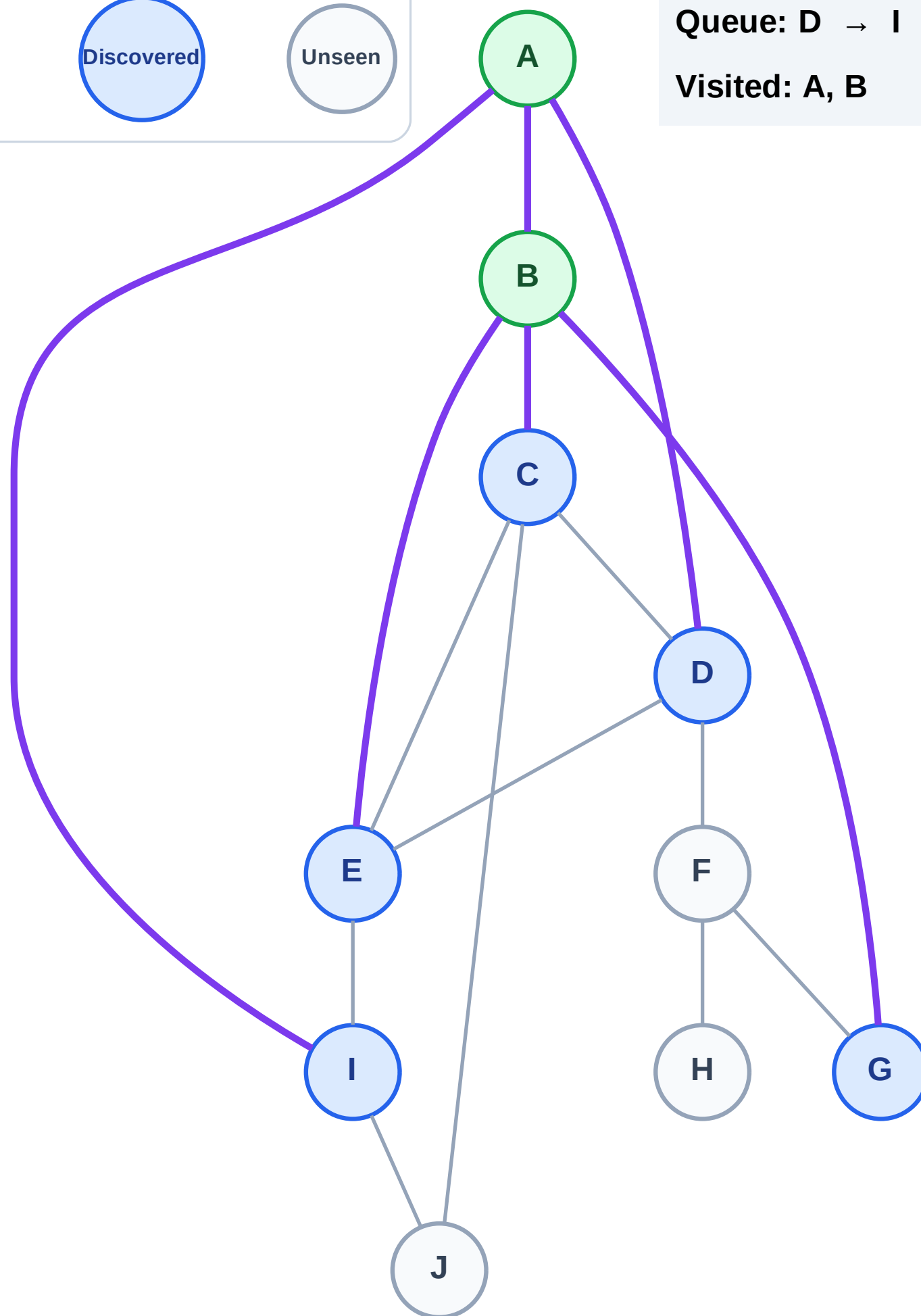
Step 5: Discover C, E, G from B

Legend



Queue: D → I → C → E → G

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

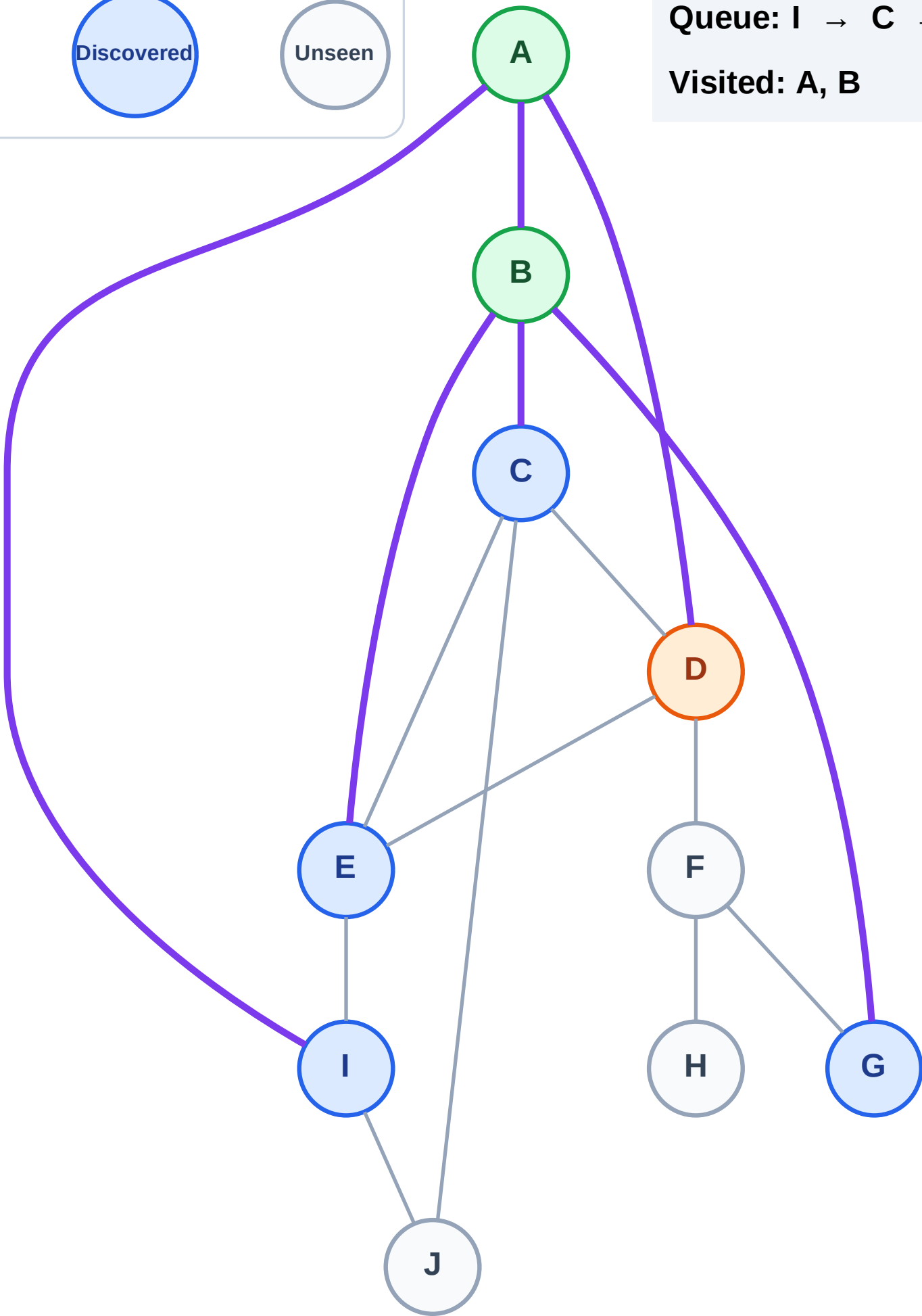
Processing

Discovered

Unseen

Queue: I → C → E → G

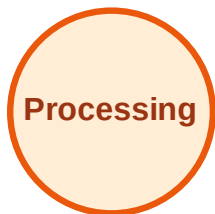
Visited: A, B



BFS Traversal

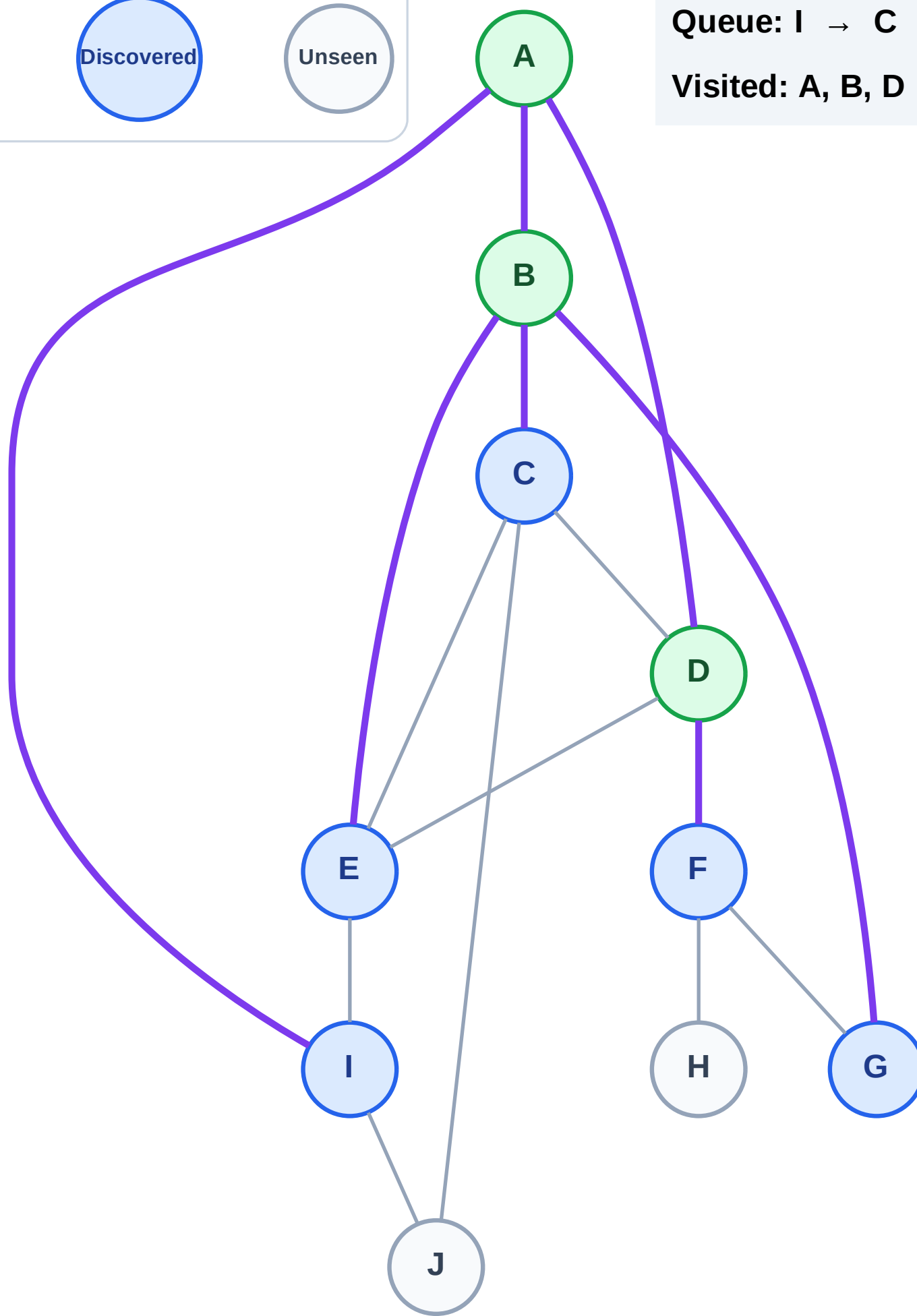
Step 7: Discover F from D

Legend



Queue: I → C → E → G → F

Visited: A, B, D



BFS Traversal

Step 8: Process node I

Legend

Complete

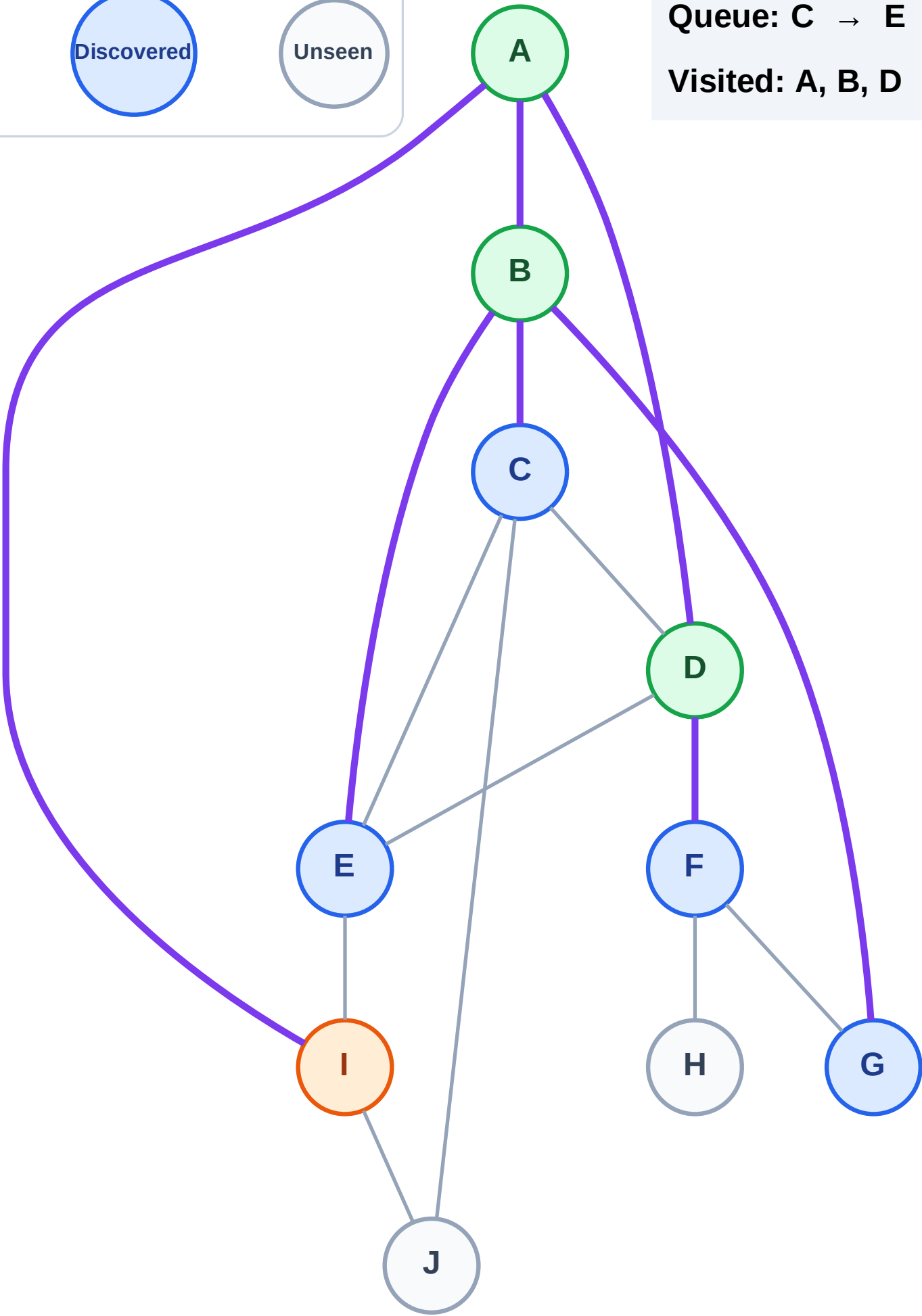
Processing

Discovered

Unseen

Queue: C → E → G → F

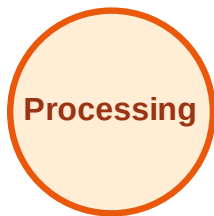
Visited: A, B, D



BFS Traversal

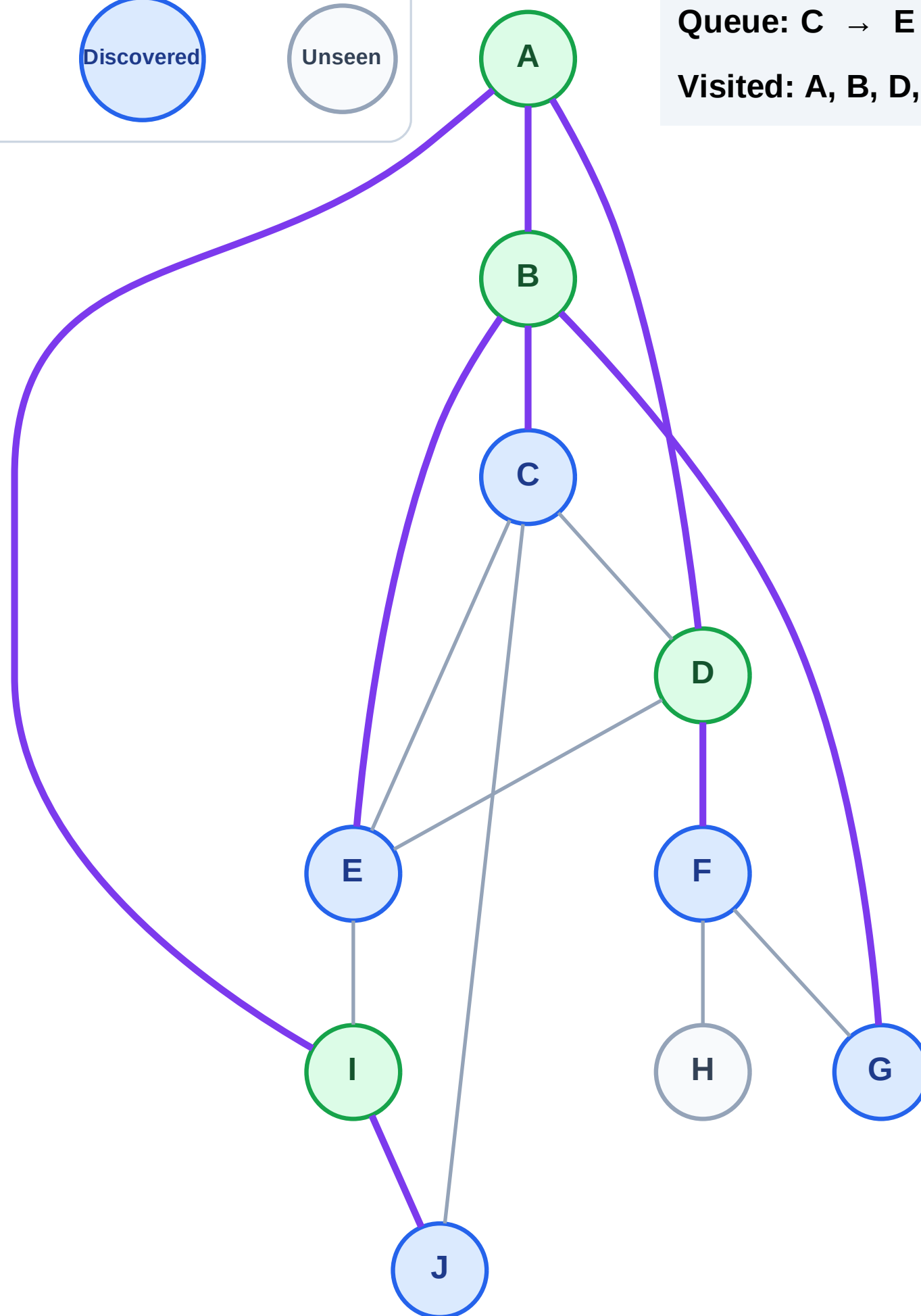
Step 9: Discover J from I

Legend



Queue: C → E → G → F → J

Visited: A, B, D, I



BFS Traversal

Step 10: Process node C

Legend

Complete

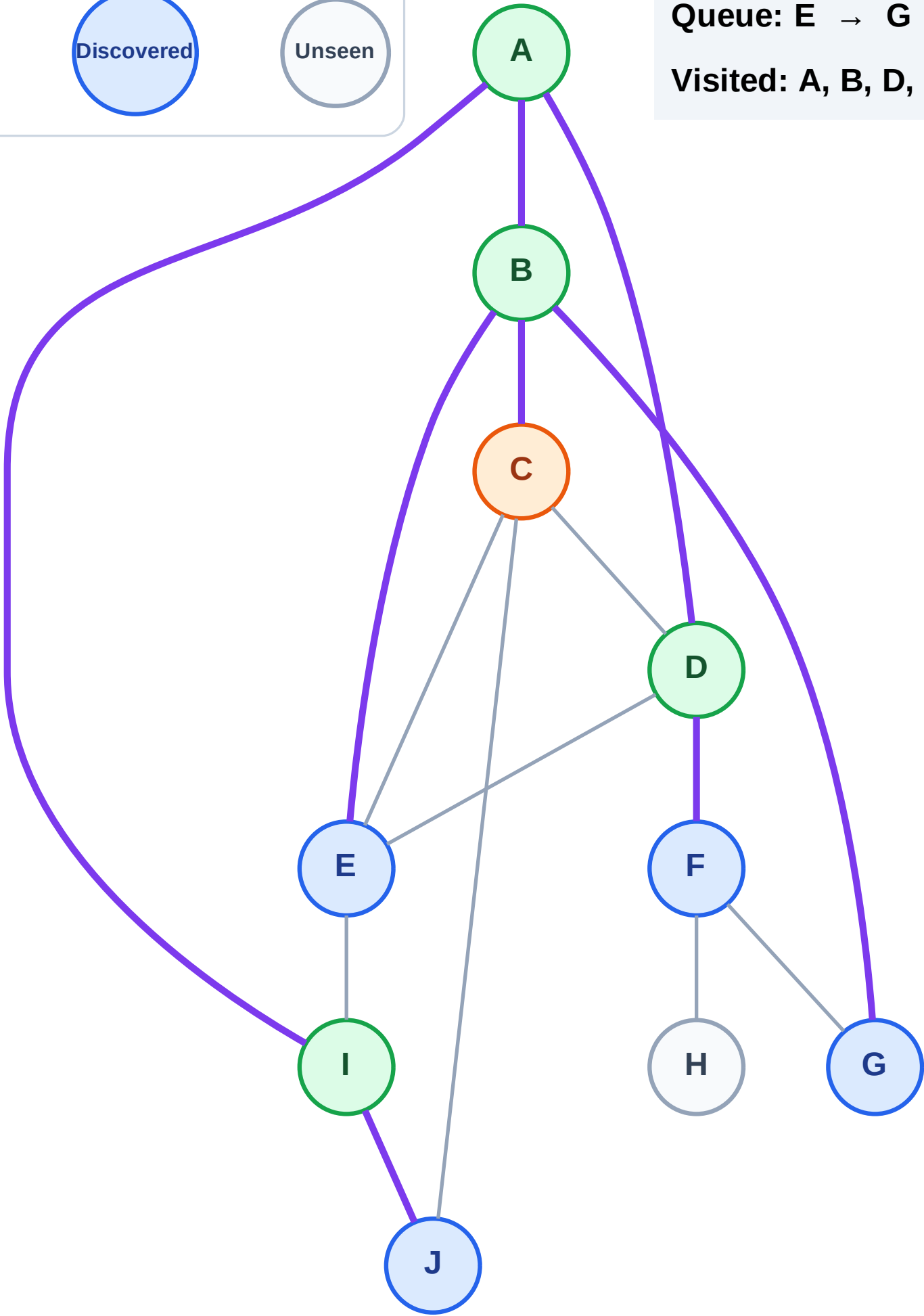
Processing

Discovered

Unseen

Queue: E → G → F → J

Visited: A, B, D, I



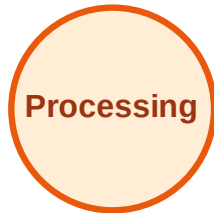
BFS Traversal

Step 11: No new neighbors from C

Legend



Complete



Processing



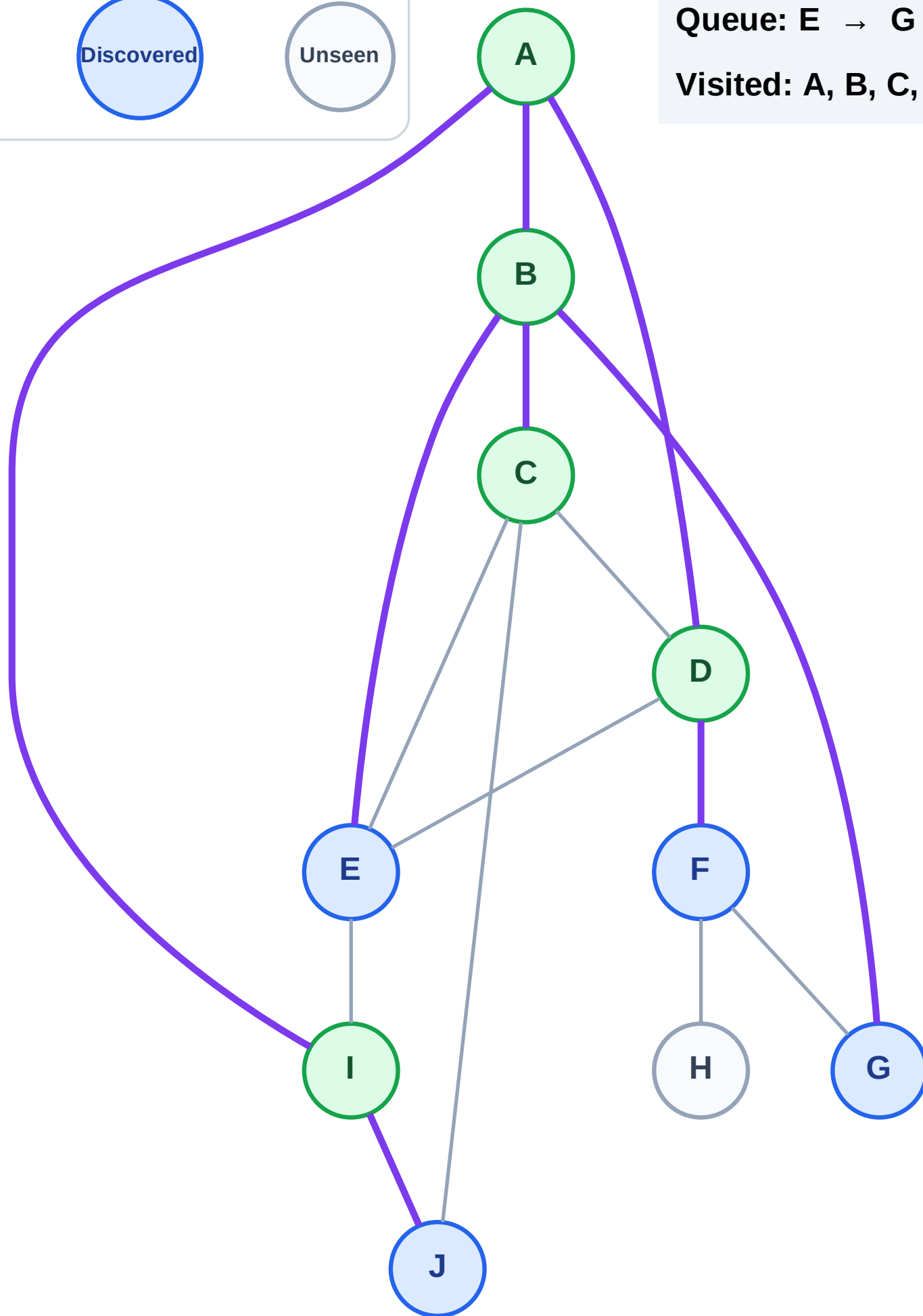
Discovered



Unseen

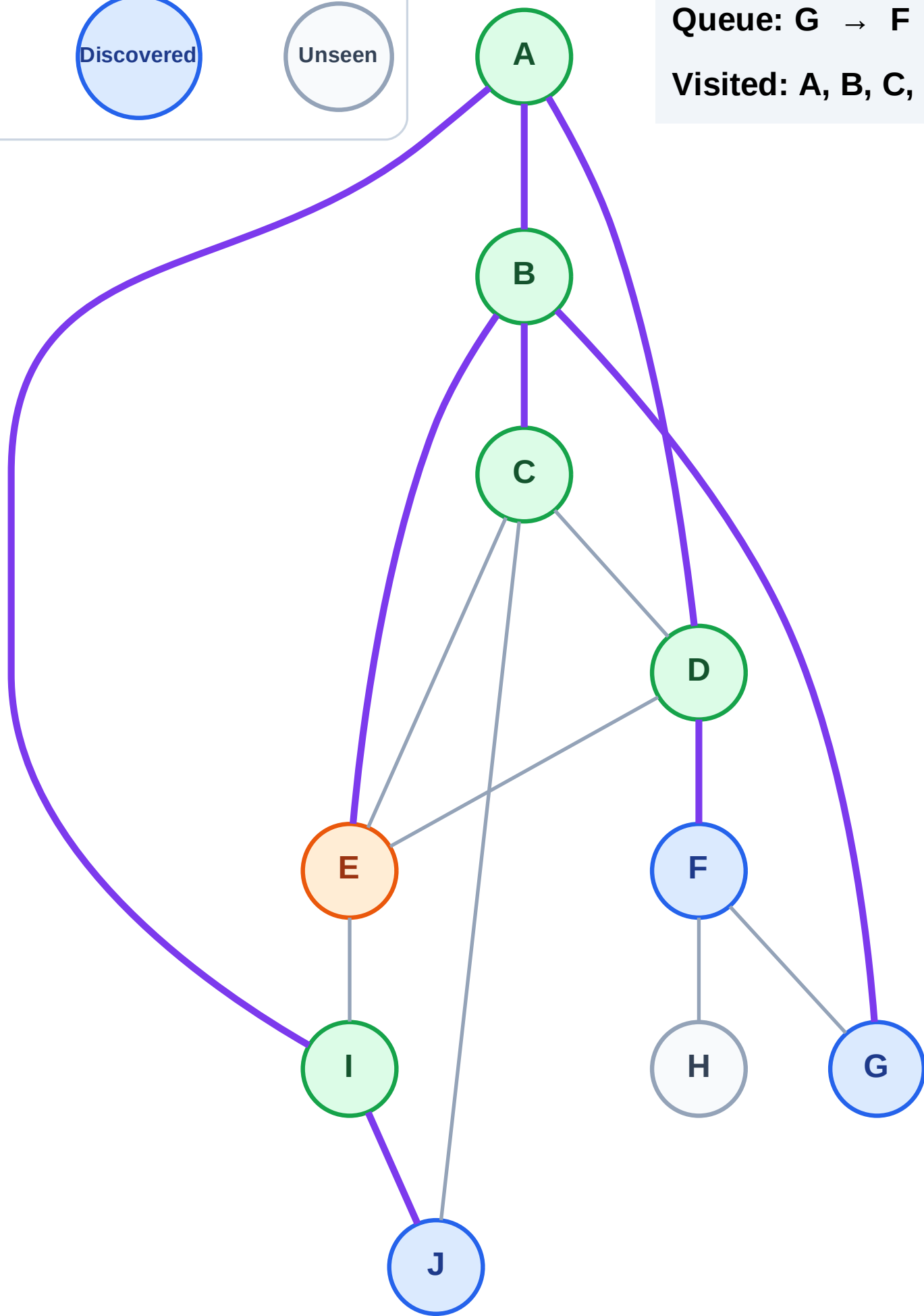
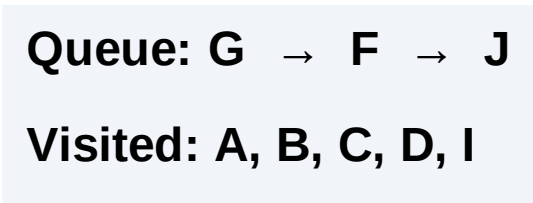
Queue: E → G → F → J

Visited: A, B, C, D, I



Step 12: Process node E

Page 10 of 10

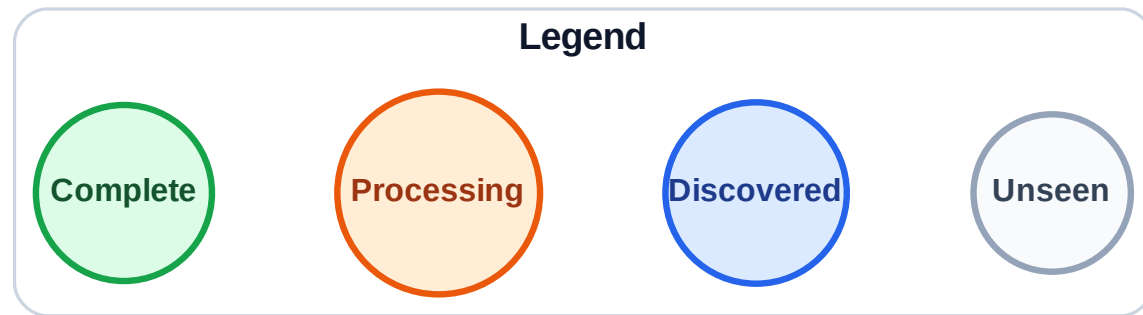


BFS Traversal

Step 13: No new neighbors from E

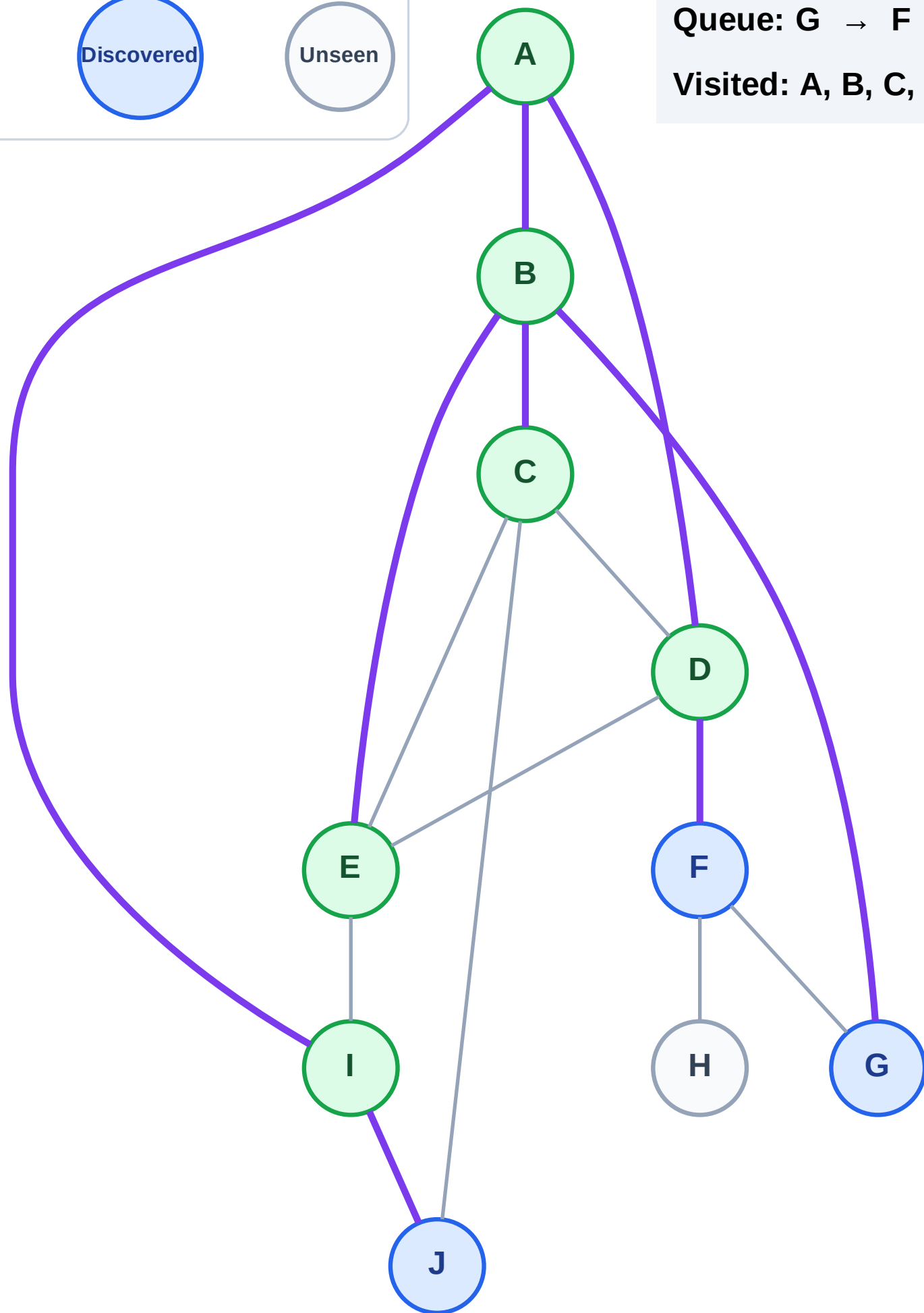
BFS Traversal

Step 13: No new neighbors from E



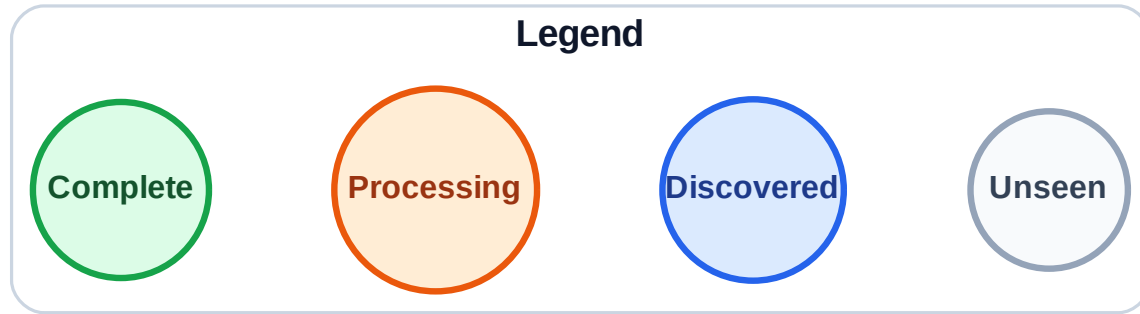
Queue: G → F → J

Visited: A, B, C, D, E, I

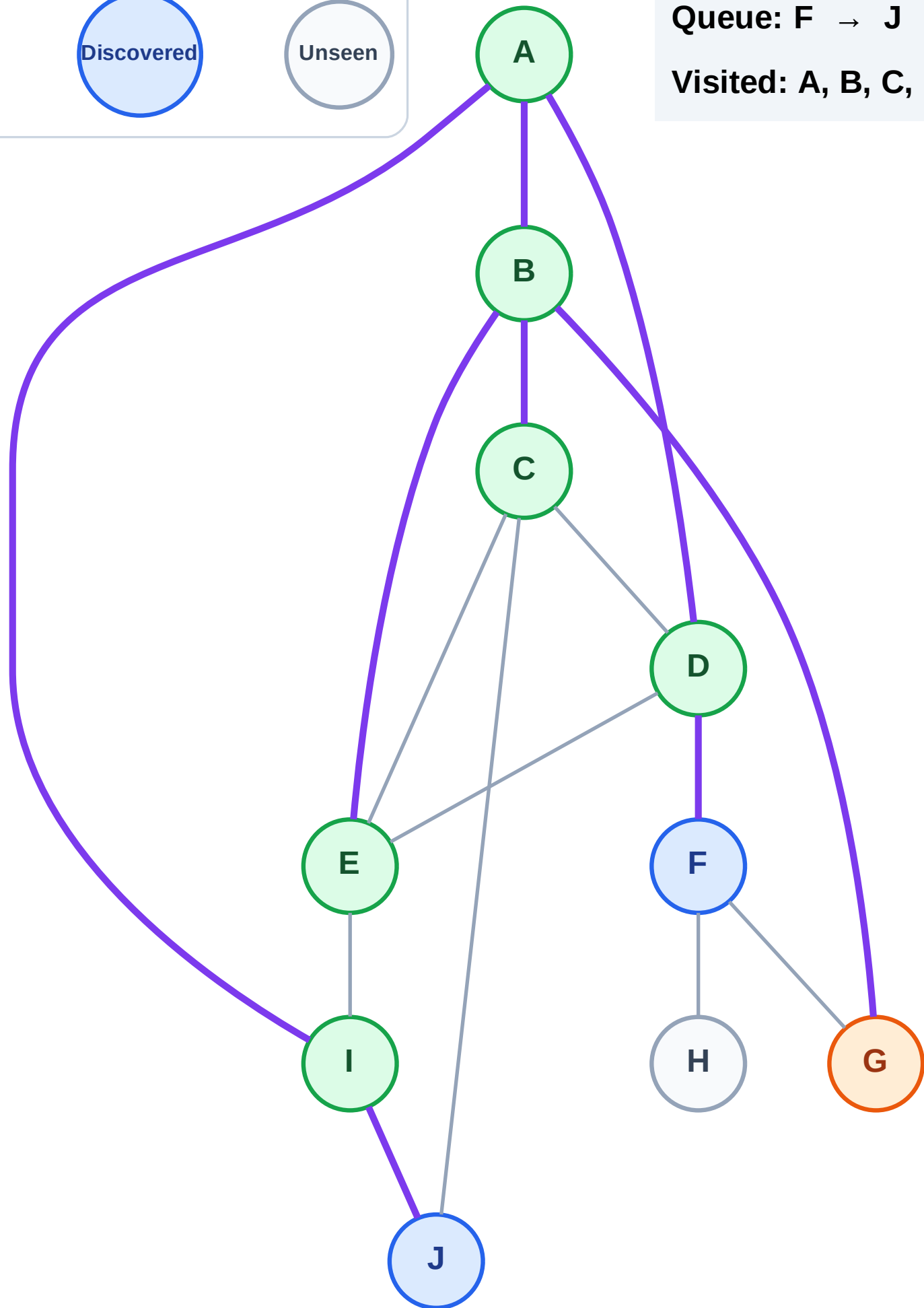


BFS Traversal

Step 14: Process node G



Queue: F → J
Visited: A, B, C, D, E, I

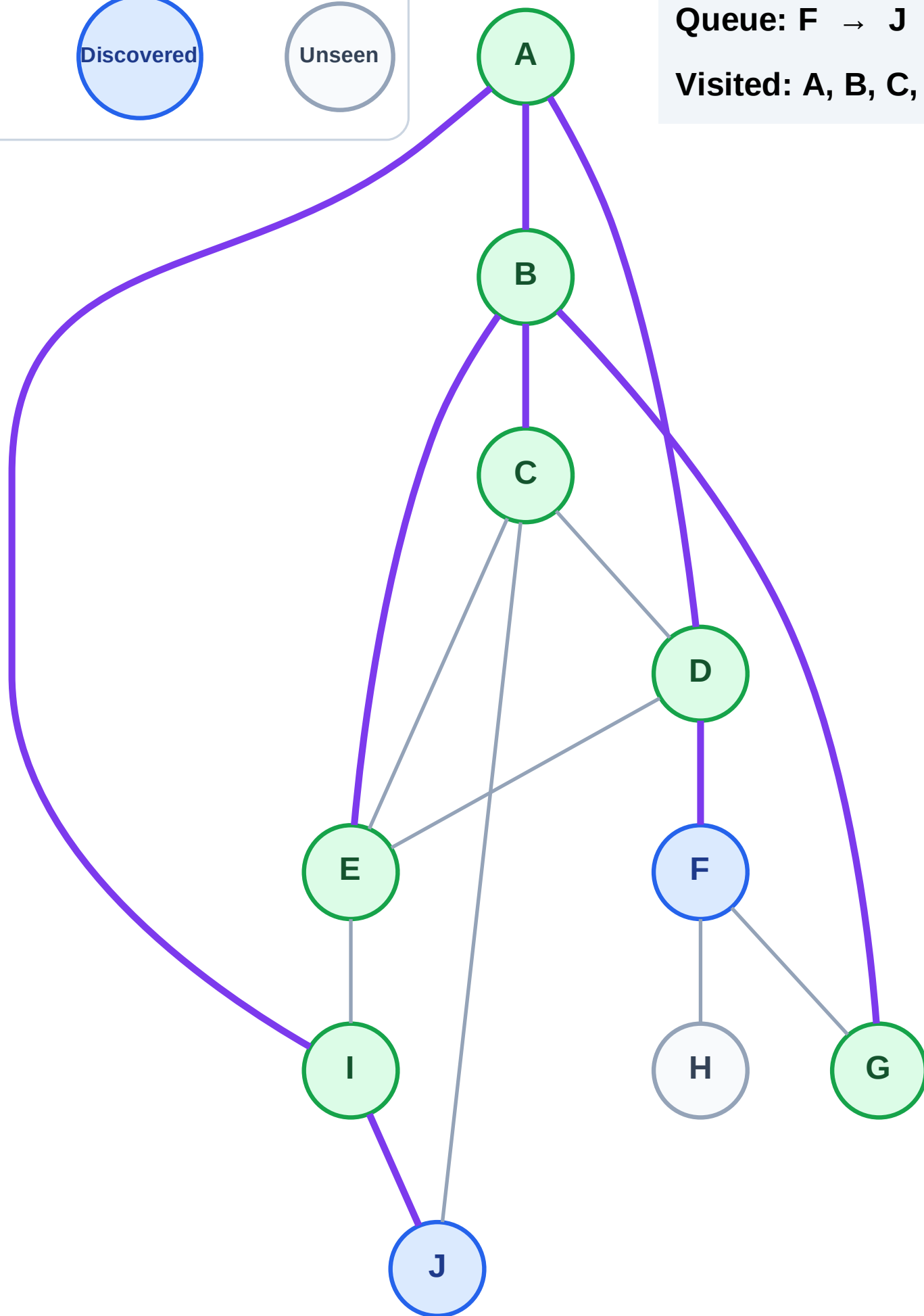


Step 15: No new neighbors from G

Legend

Unseen

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node F

Legend

Complete

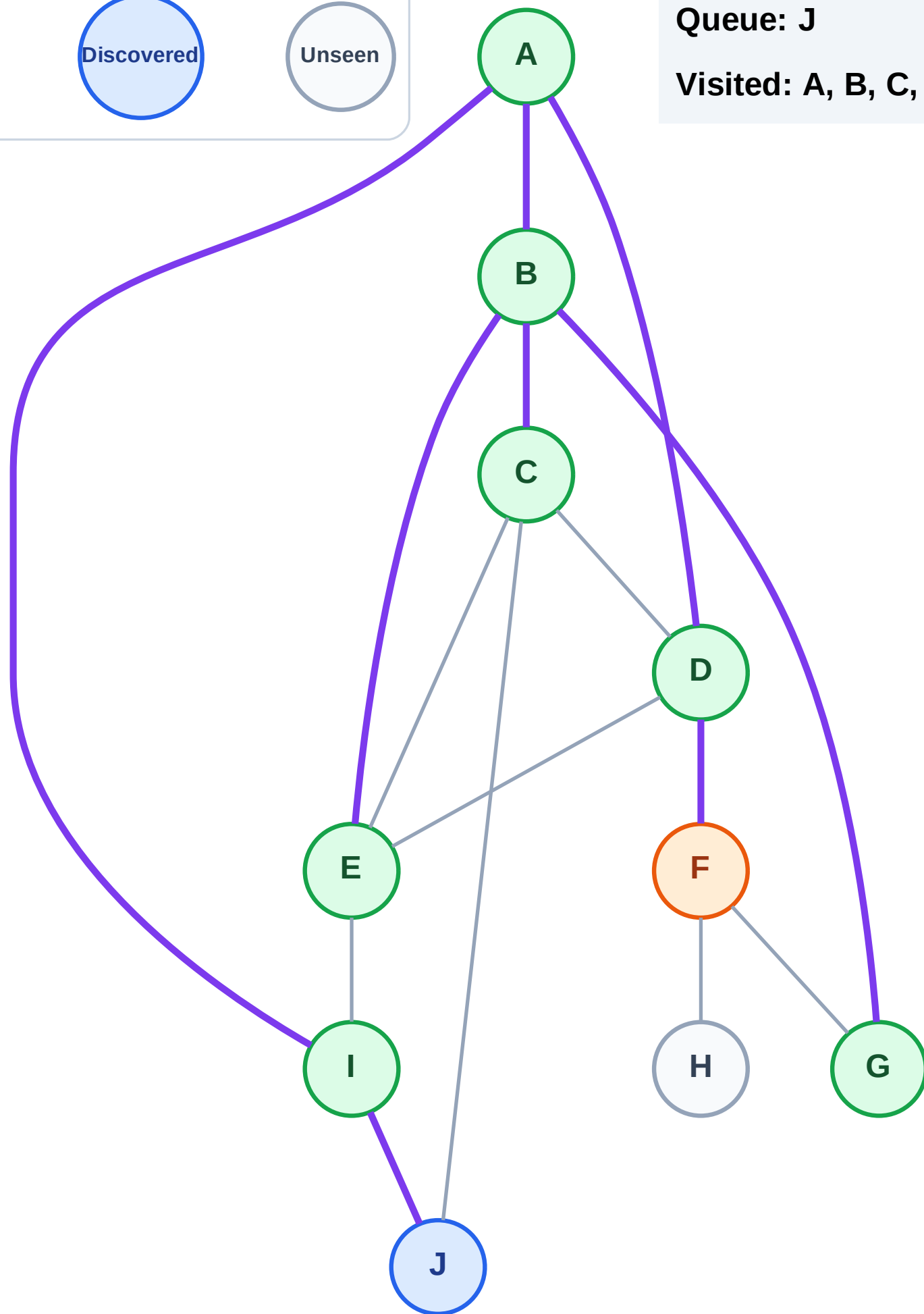
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, G, I



BFS Traversal

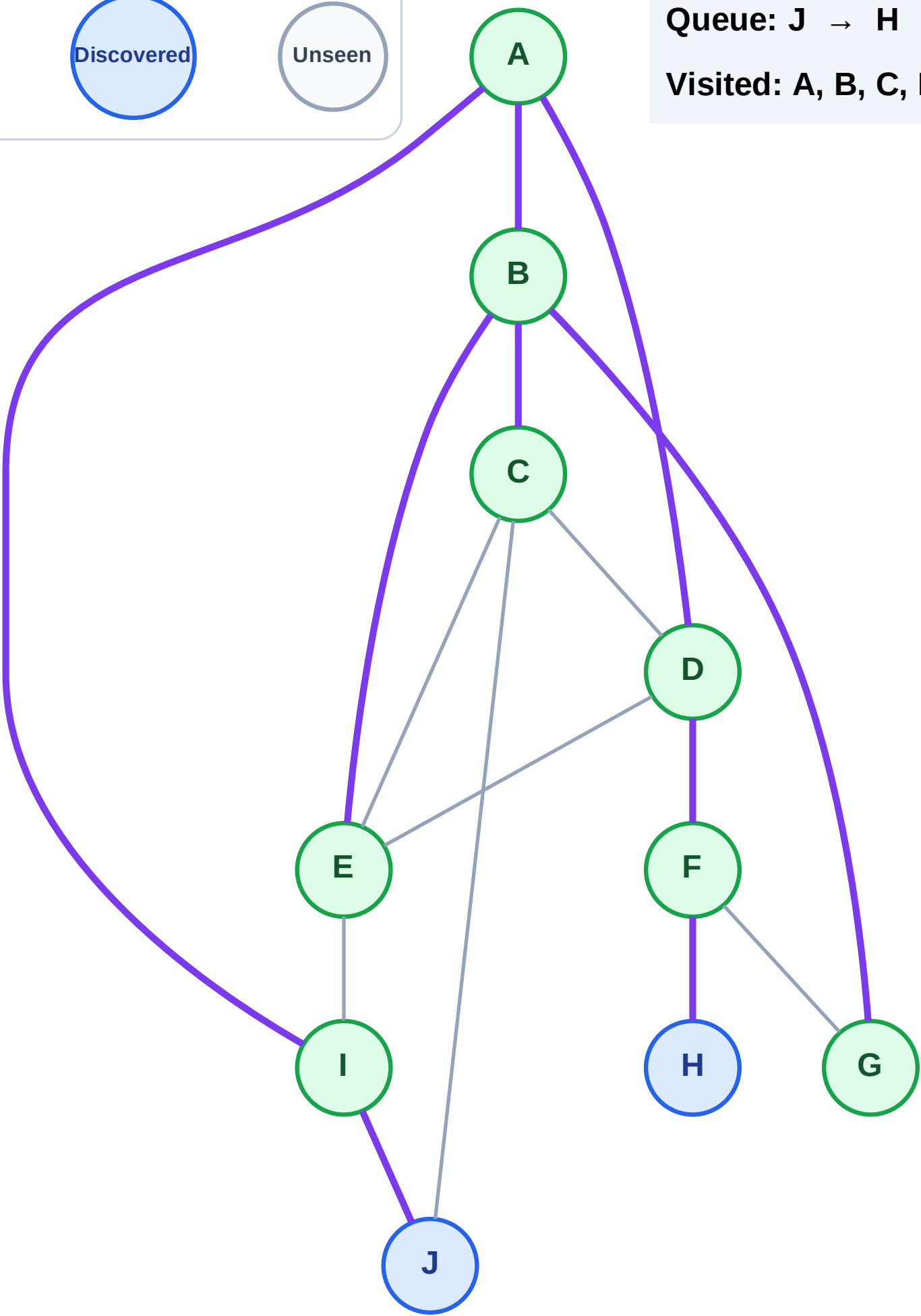
Step 17: Discover H from F

Legend



Queue: J → H

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node J

Legend

Complete

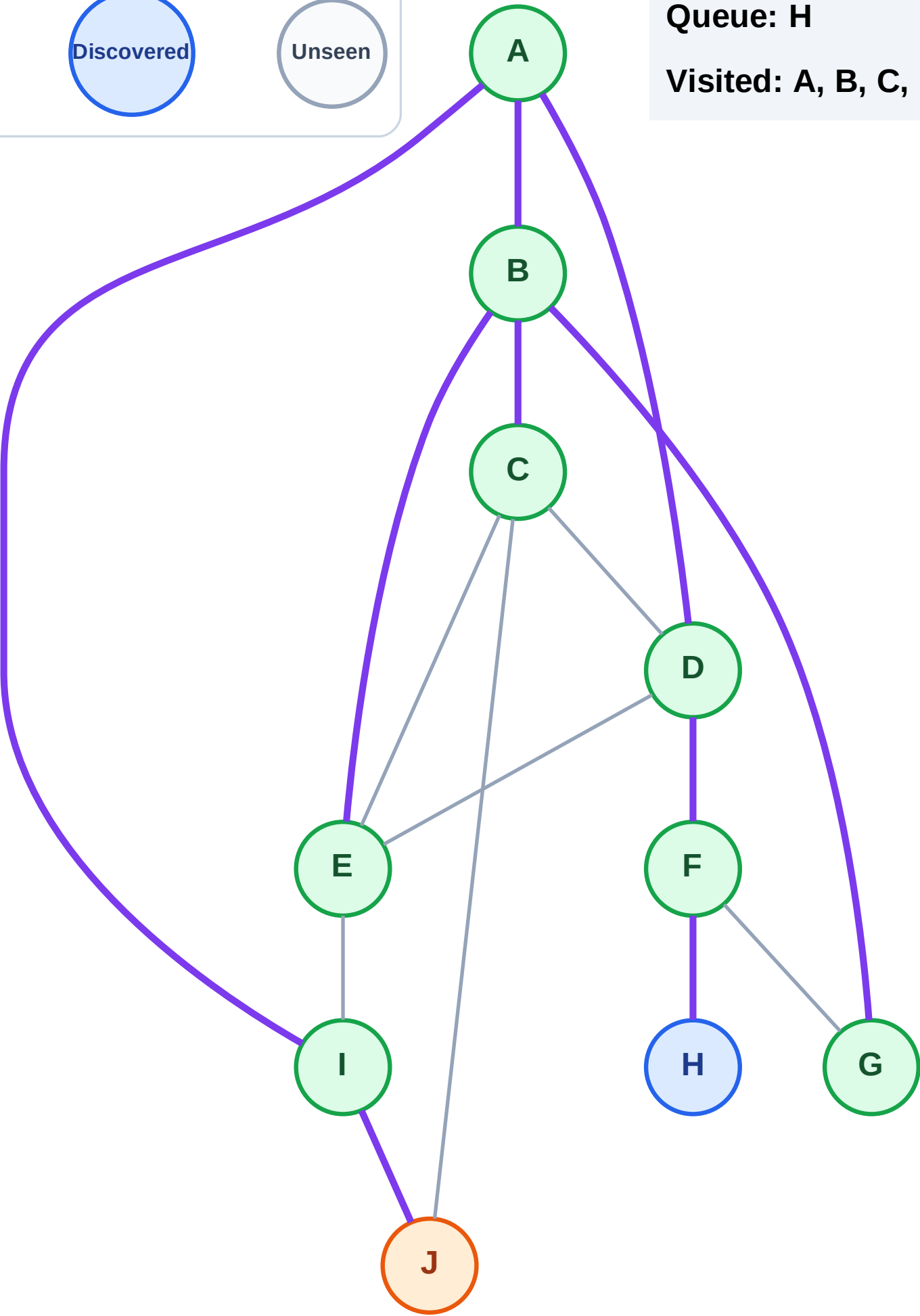
Processing

Discovered

Unseen

Queue: H

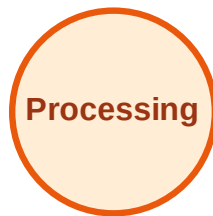
Visited: A, B, C, D, E, F, G, I



BFS Traversal

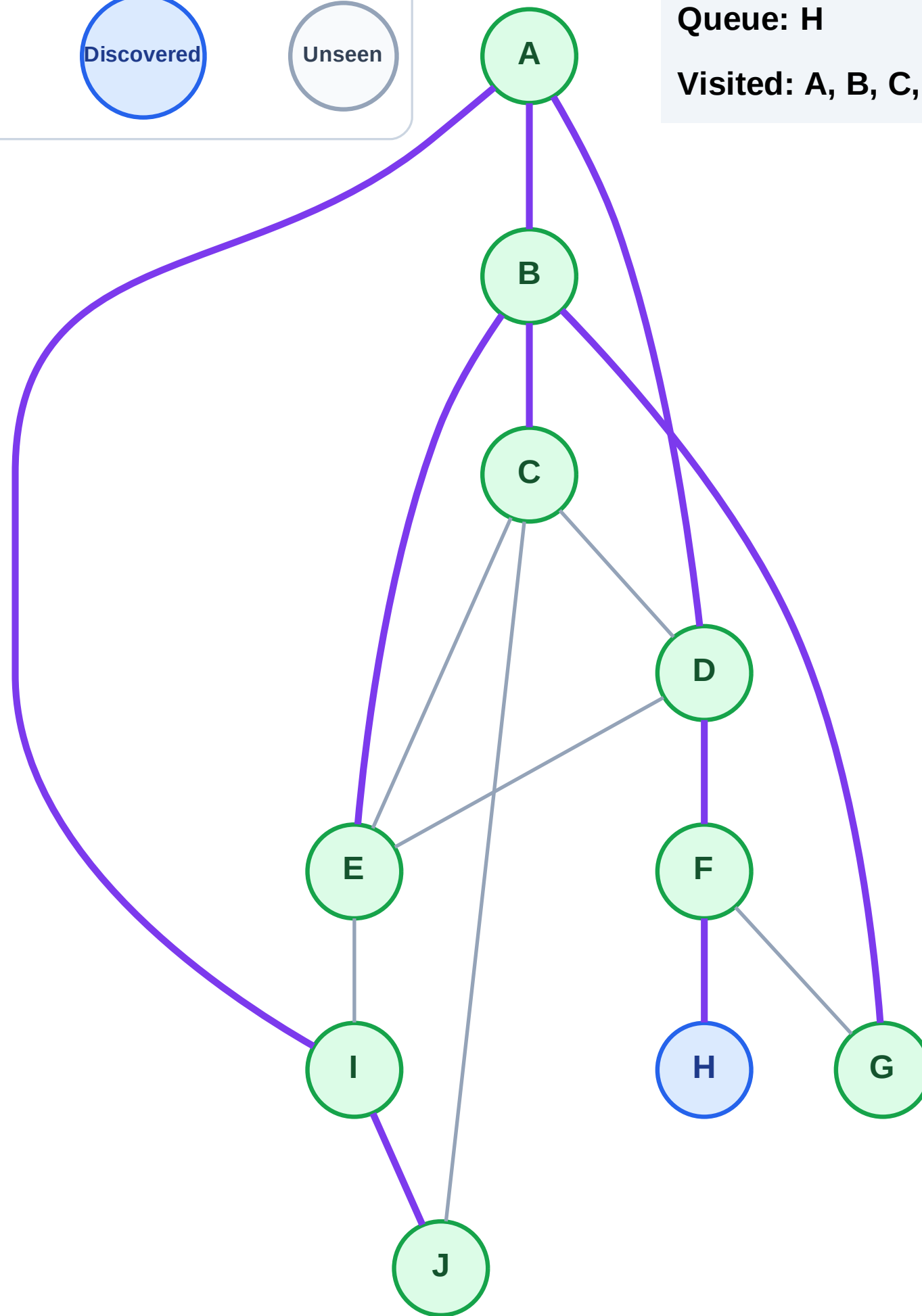
Step 19: No new neighbors from J

Legend



Queue: H

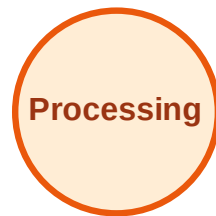
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

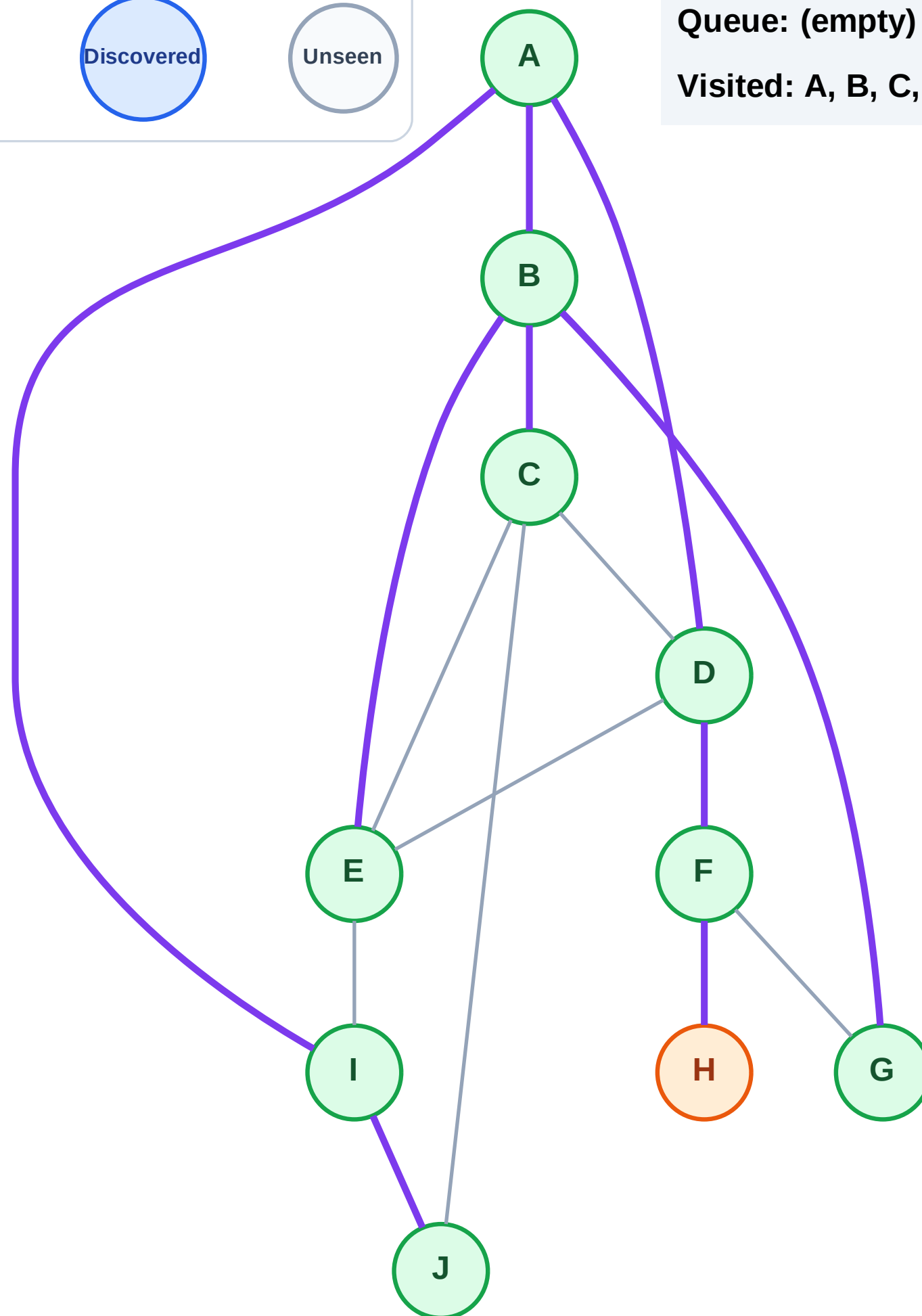
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

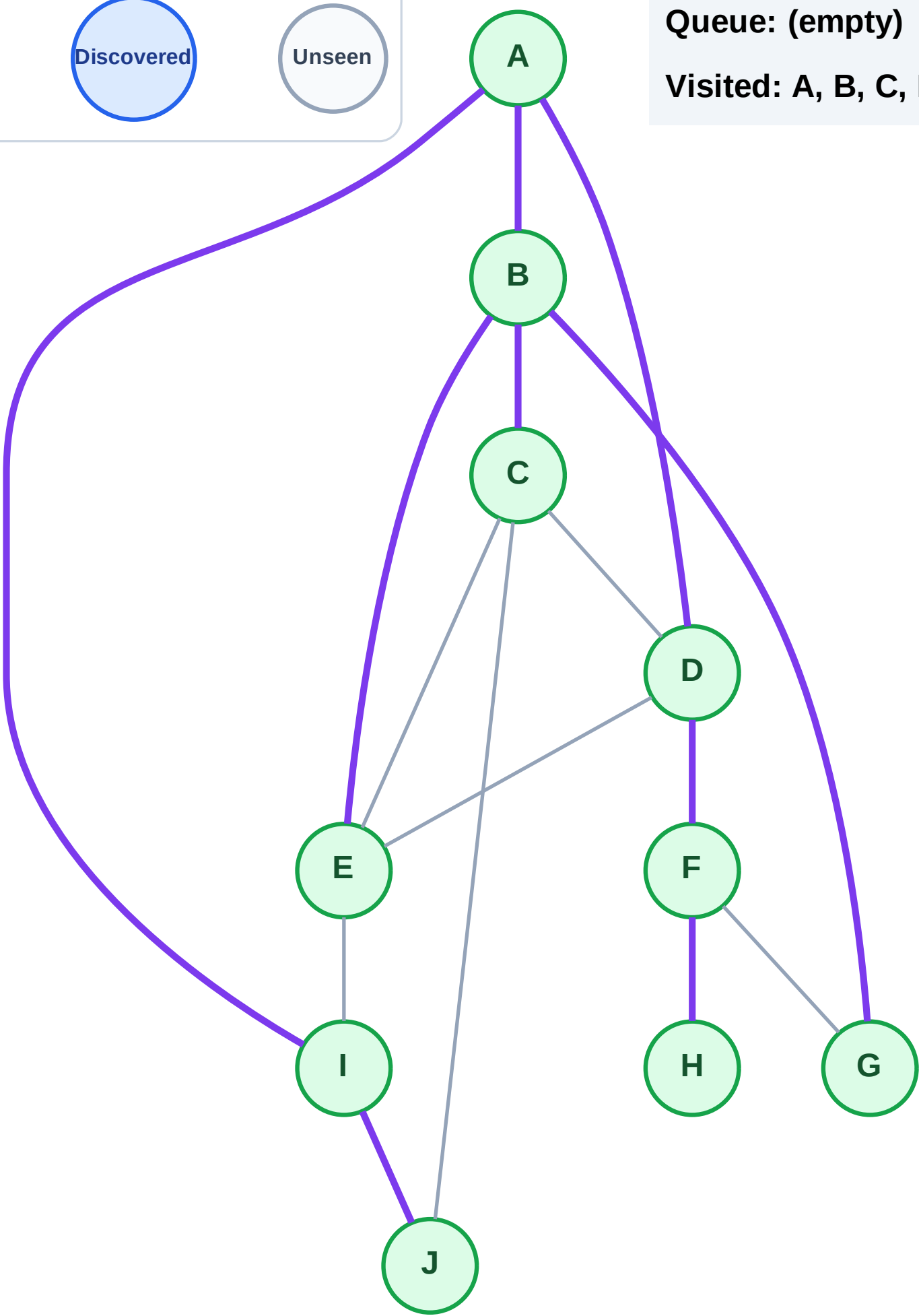
Step 21: No new neighbors from H

Legend



Queue: (empty)

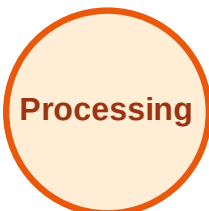
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

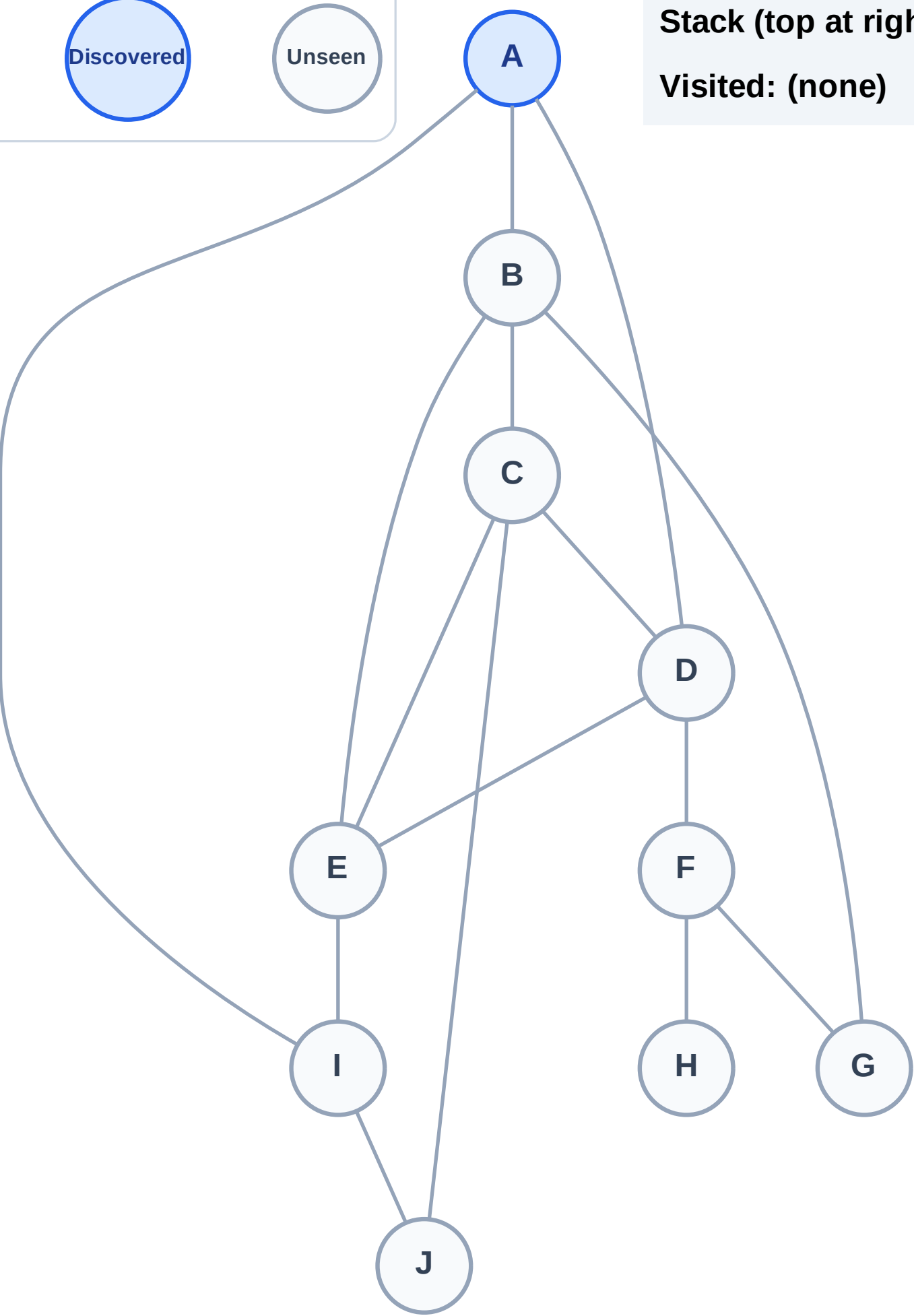
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

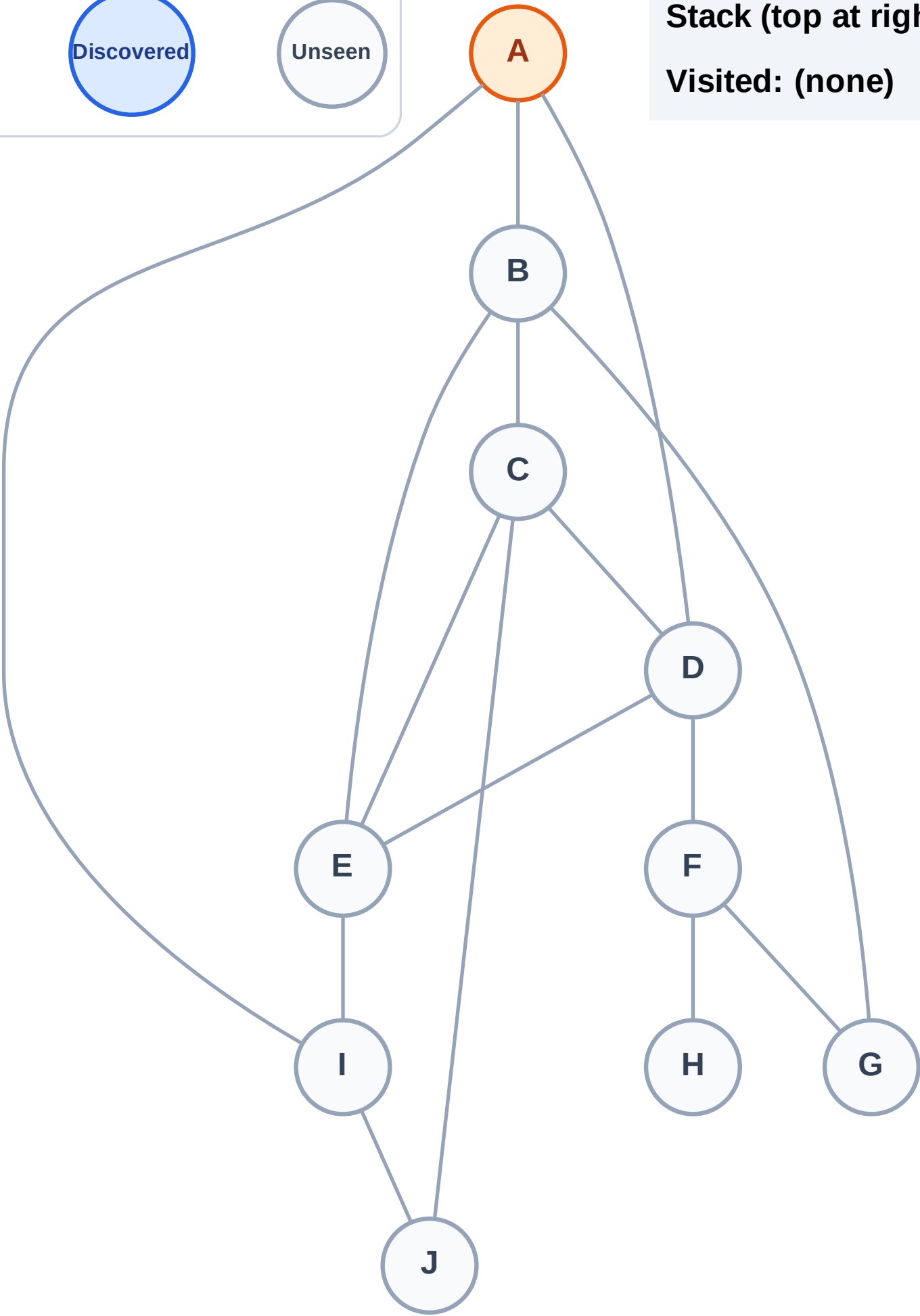
Complete

Processing

Discovered

Unseen

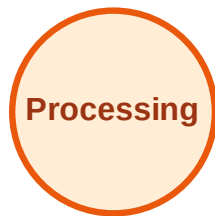
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

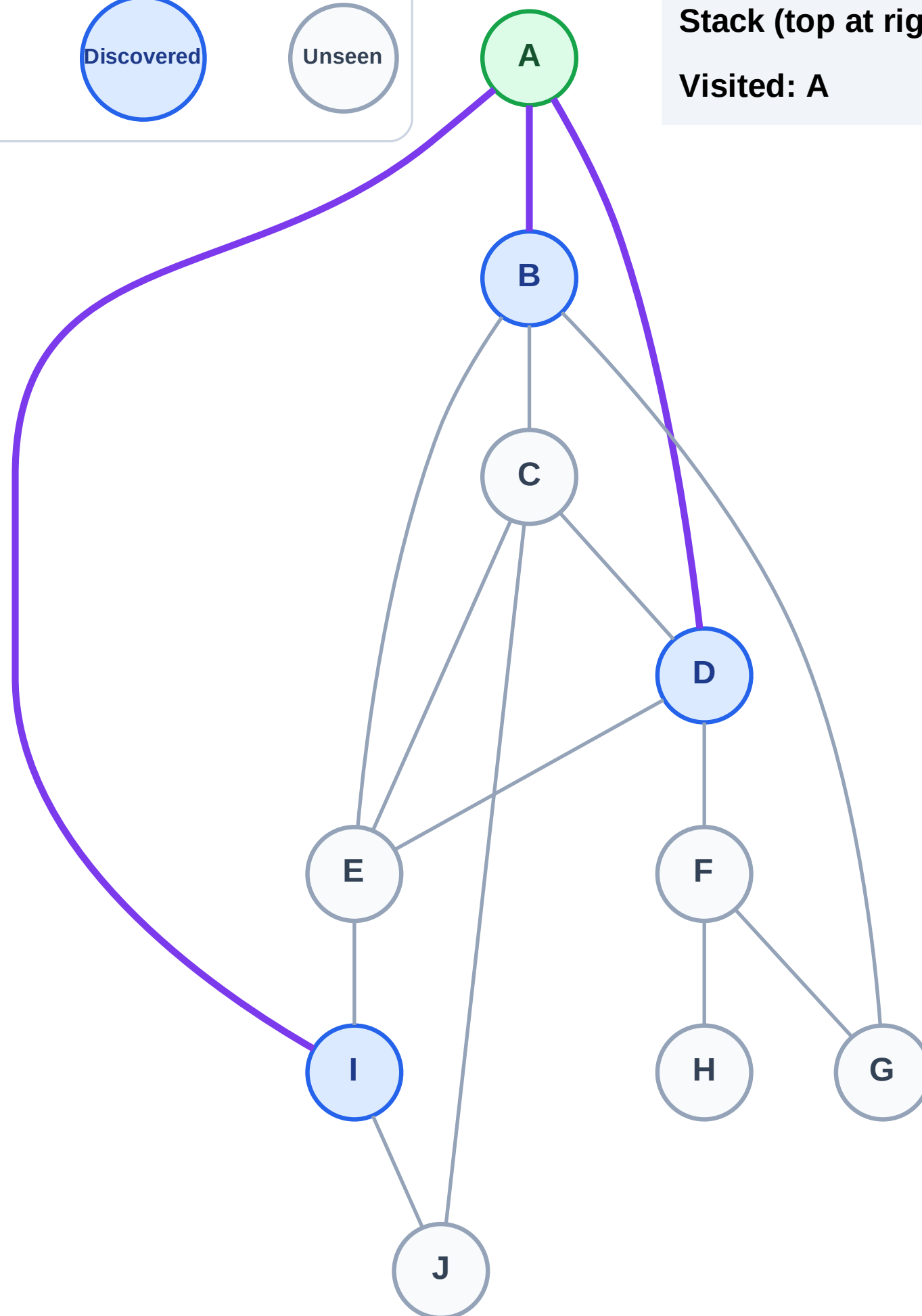
Step 3: Discover I, D, B from A

Legend



Stack (top at right): I | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

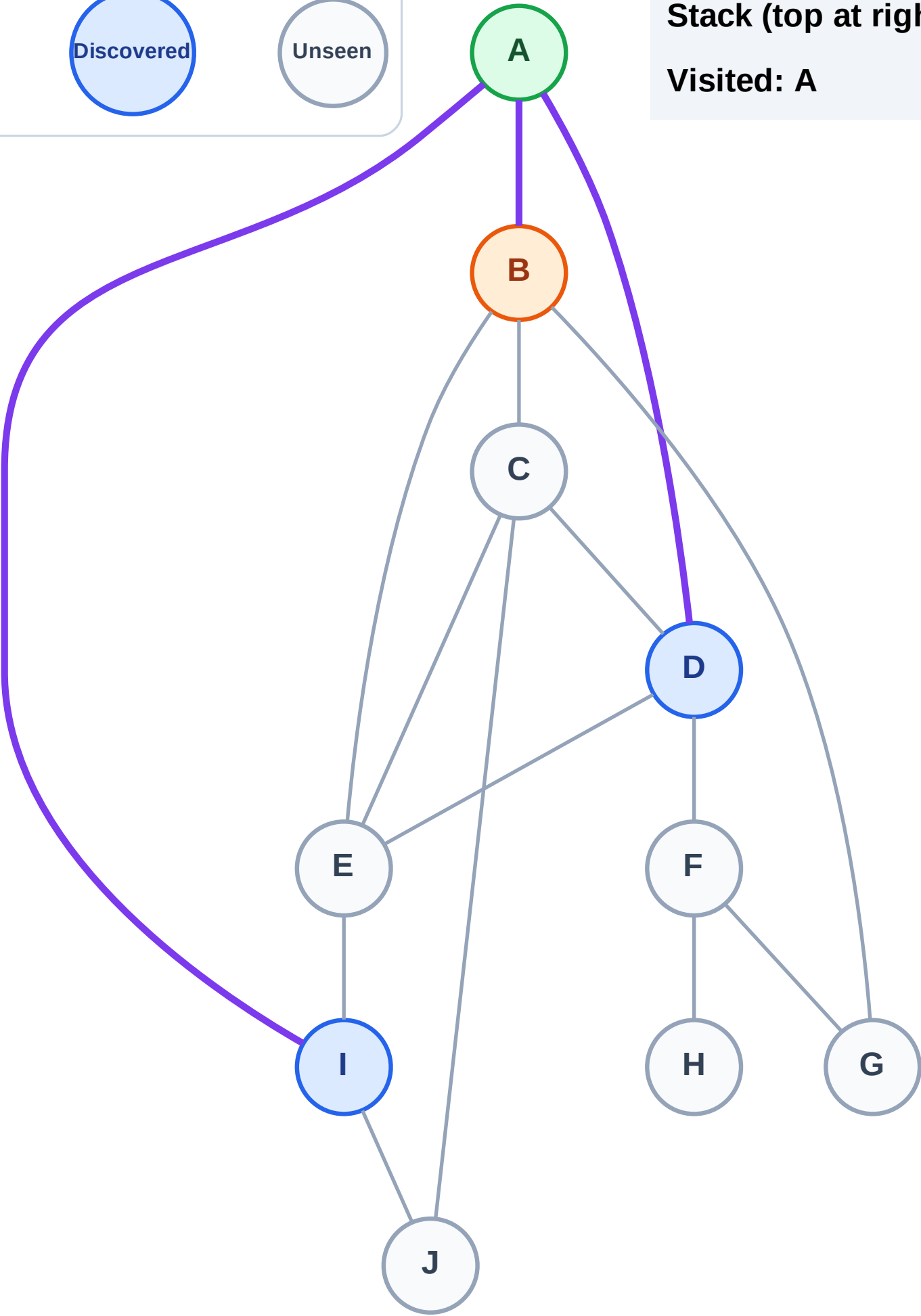
Processing

Discovered

Unseen

Stack (top at right): I | D

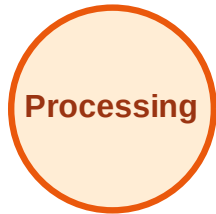
Visited: A



DFS Traversal

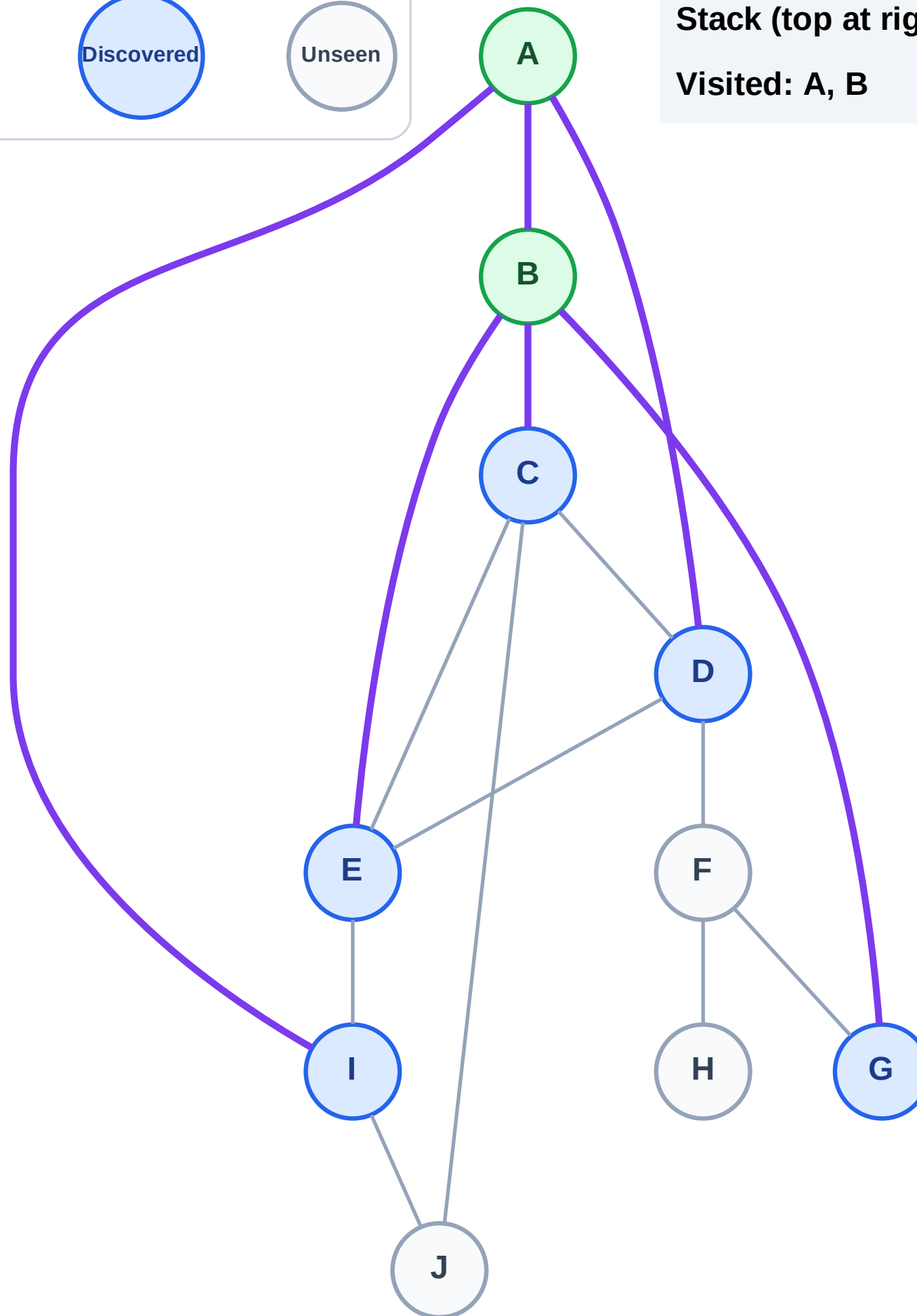
Step 5: Discover G, E, C from B

Legend



Stack (top at right): I | D | G | E | C

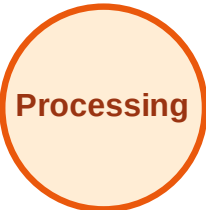
Visited: A, B



DFS Traversal

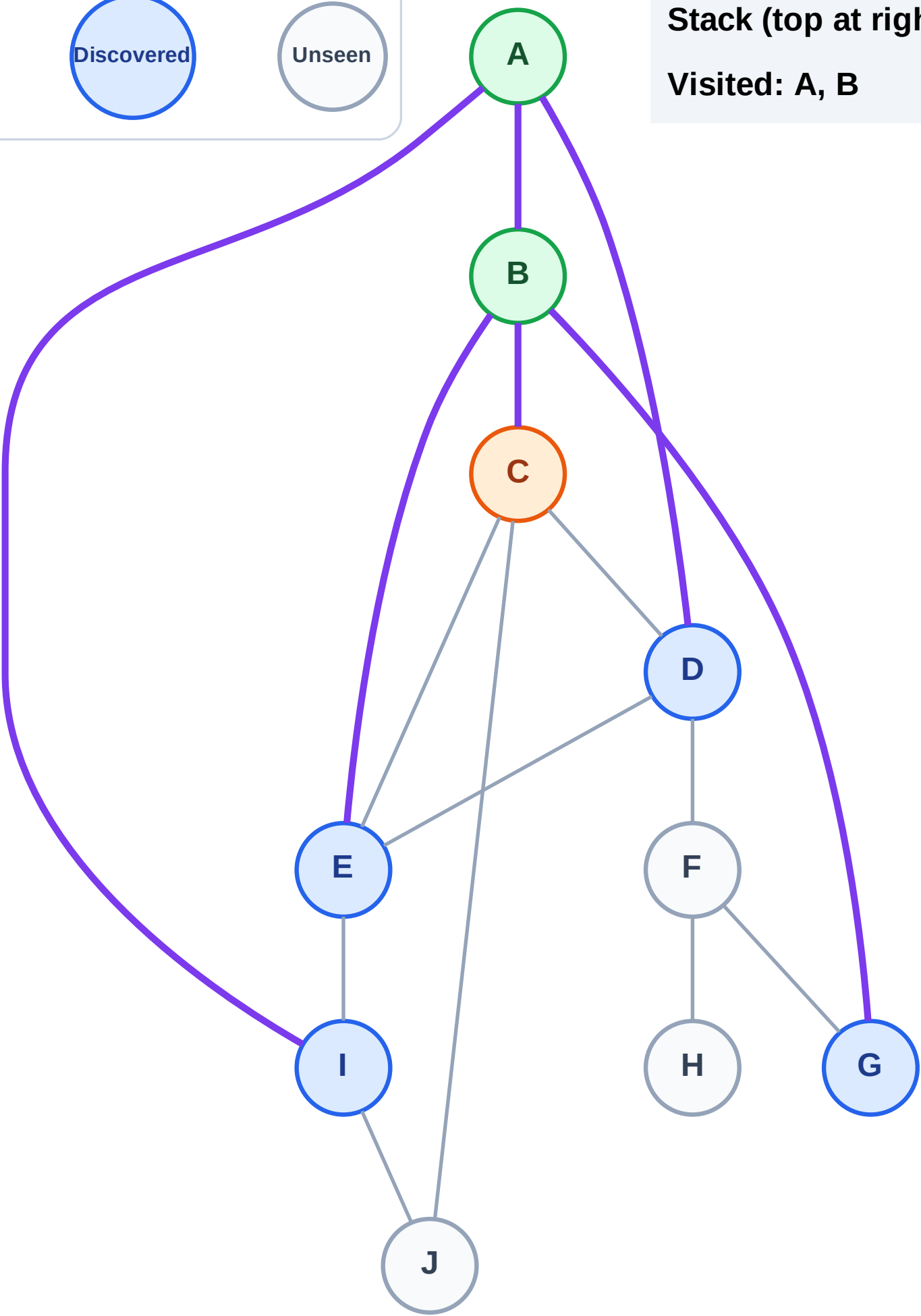
Step 6: Process node C

Legend



Stack (top at right): I | D | G | E

Visited: A, B



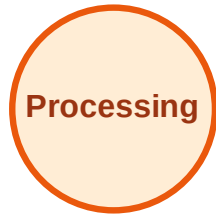
DFS Traversal

Step 7: Discover J from C

Legend



Complete



Processing



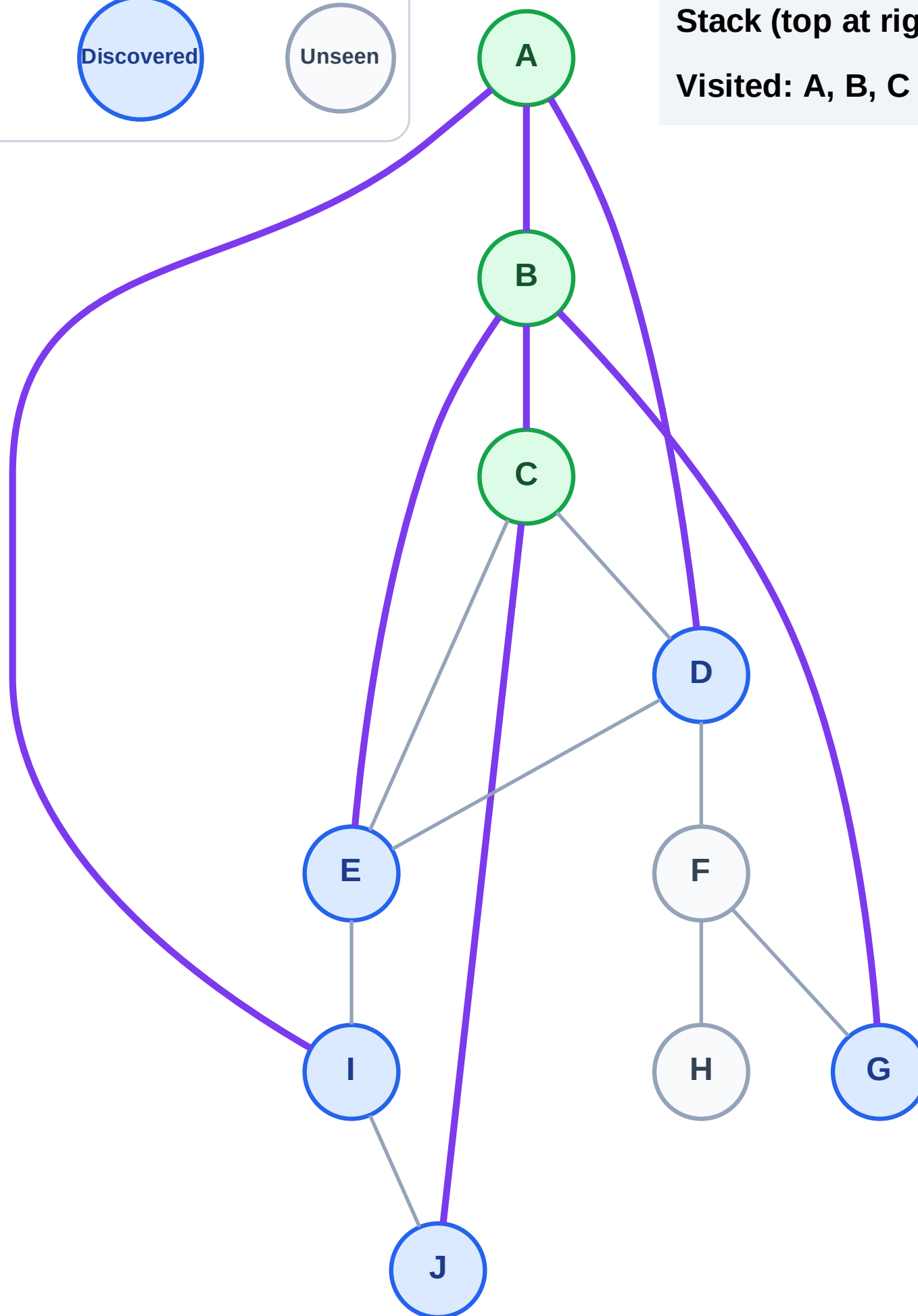
Discovered



Unseen

Stack (top at right): I | D | G | E | J

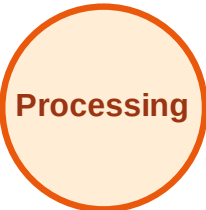
Visited: A, B, C



DFS Traversal

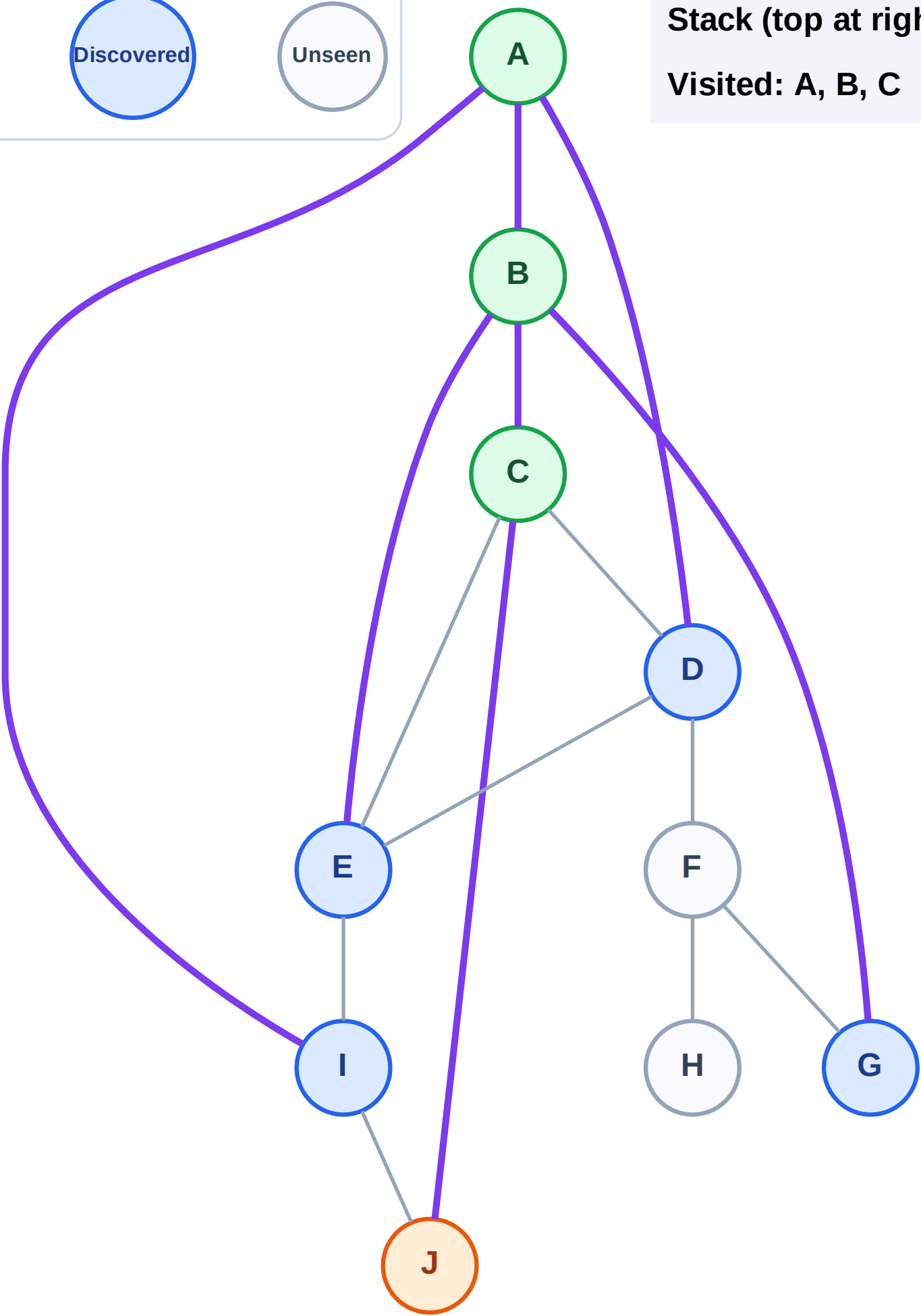
Step 8: Process node J

Legend



Stack (top at right): I | D | G | E

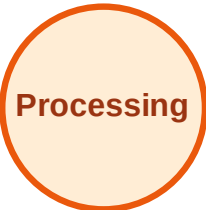
Visited: A, B, C



DFS Traversal

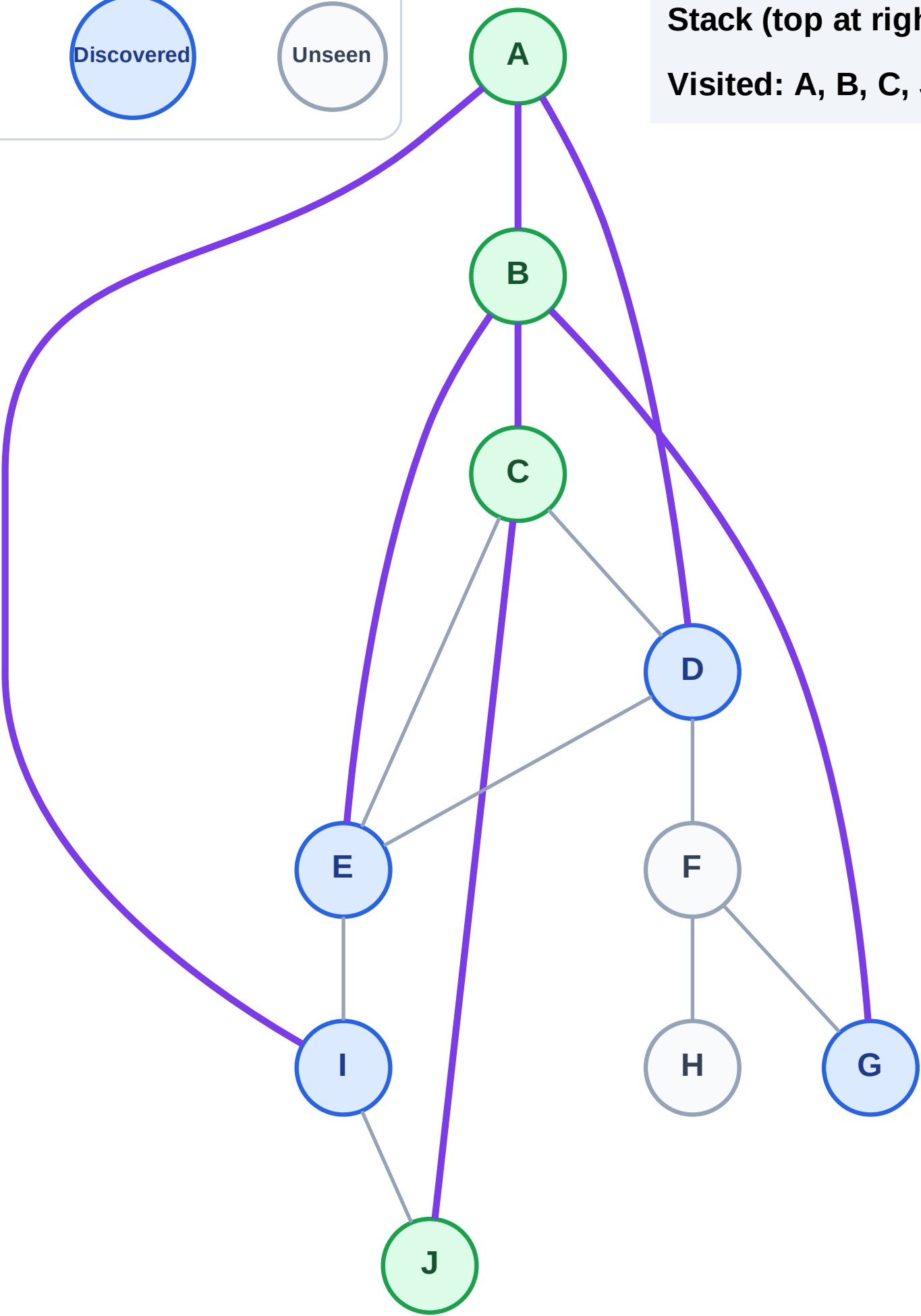
Step 9: No new neighbors from J

Legend



Stack (top at right): I | D | G | E

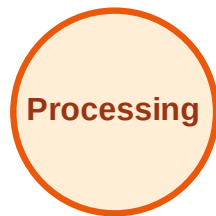
Visited: A, B, C, J



DFS Traversal

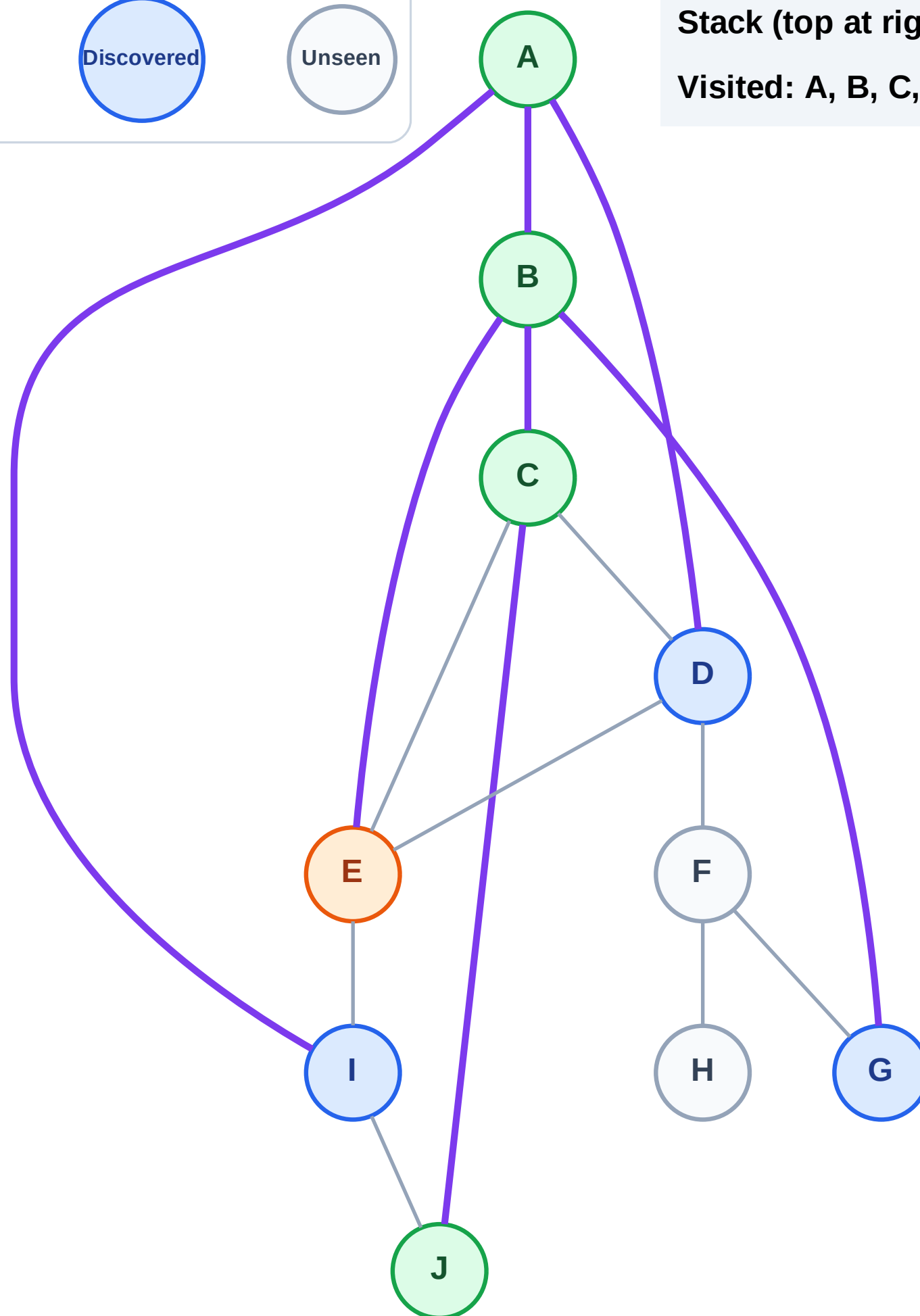
Step 10: Process node E

Legend



Stack (top at right): I | D | G

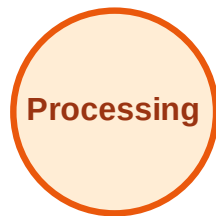
Visited: A, B, C, J



DFS Traversal

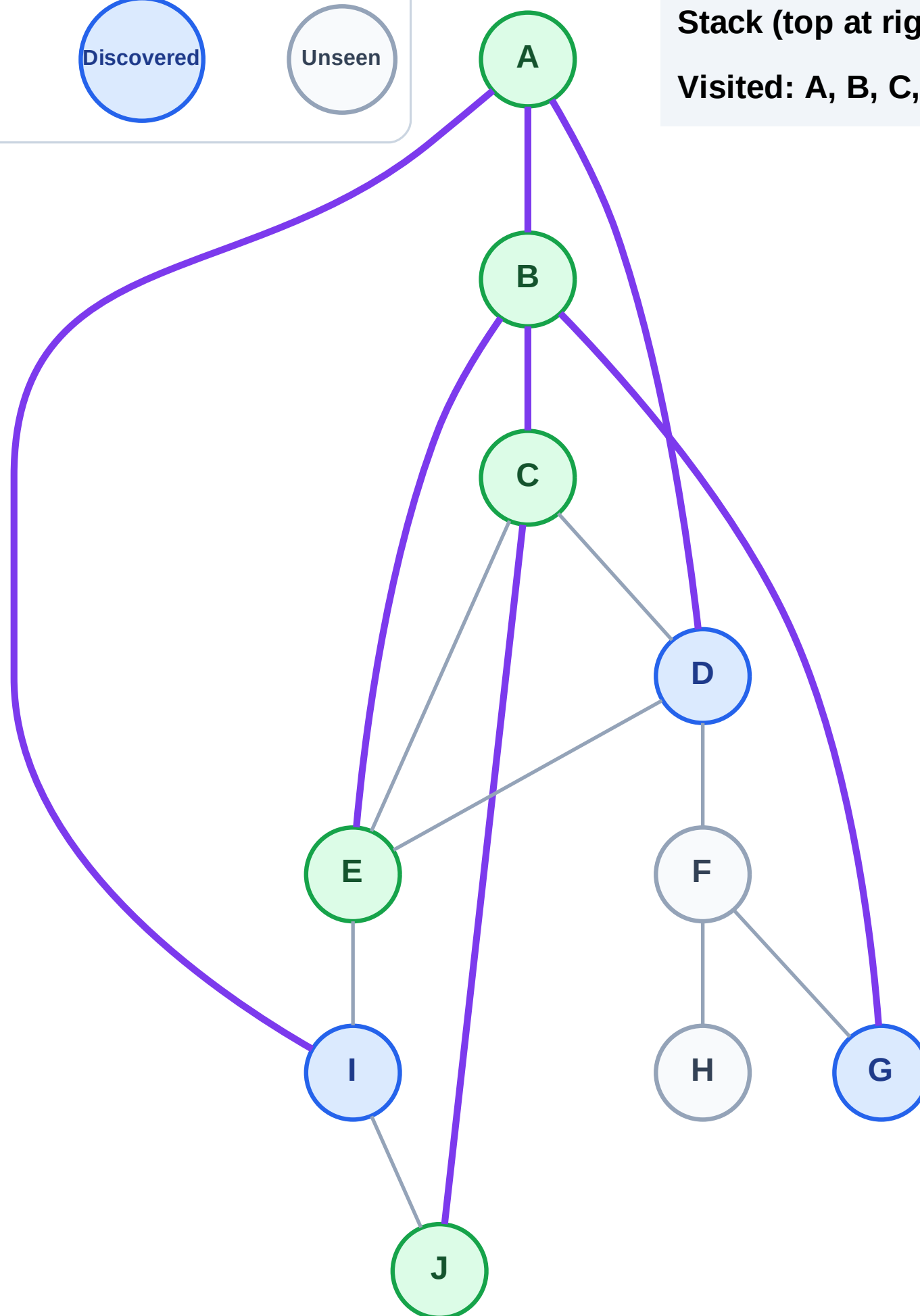
Step 11: No new neighbors from E

Legend



Stack (top at right): I | D | G

Visited: A, B, C, E, J



DFS Traversal

Step 12: Process node G

Legend

Complete

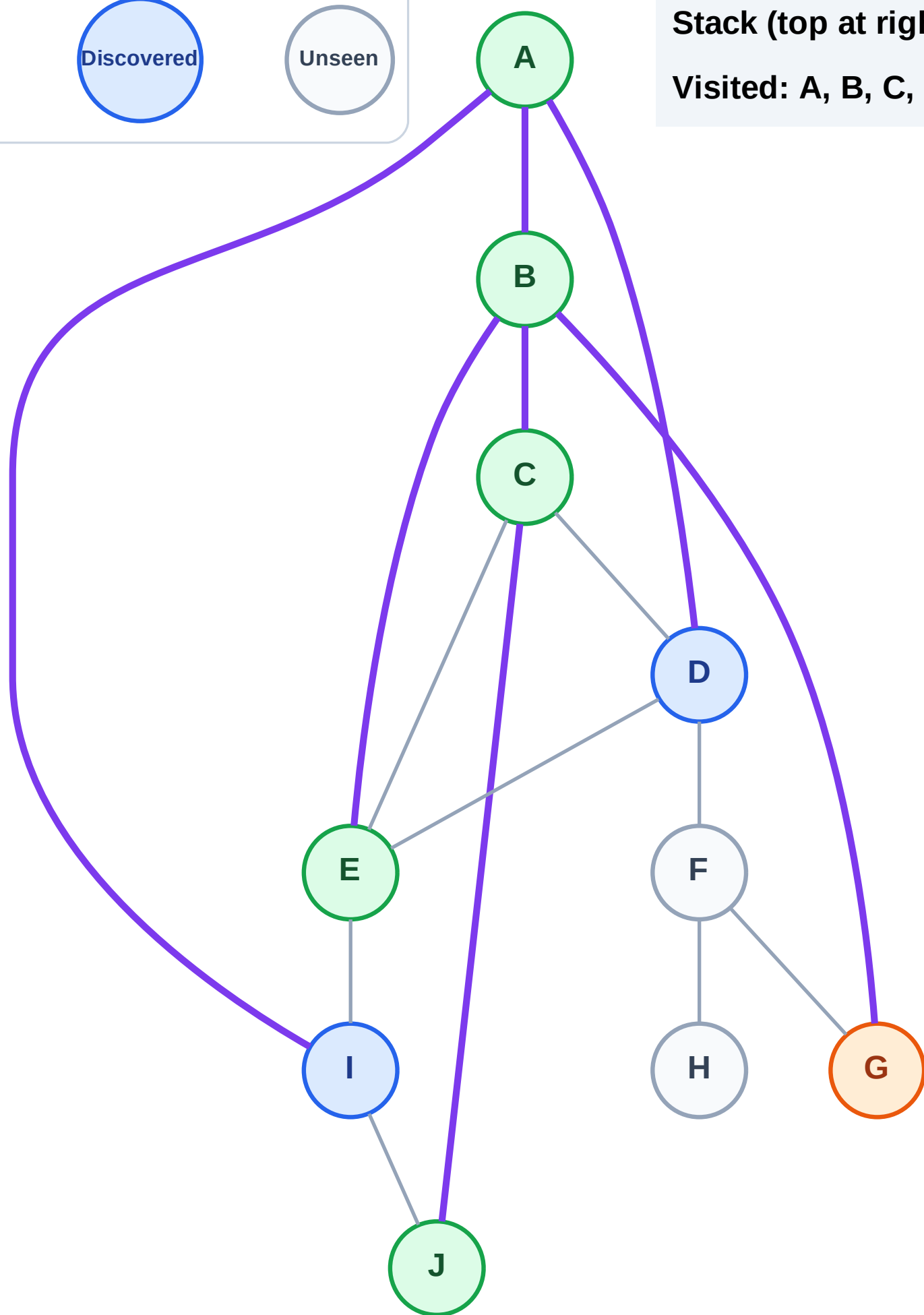
Processing

Discovered

Unseen

Stack (top at right): I | D

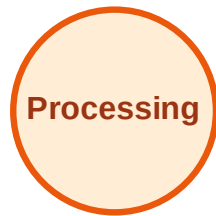
Visited: A, B, C, E, J



DFS Traversal

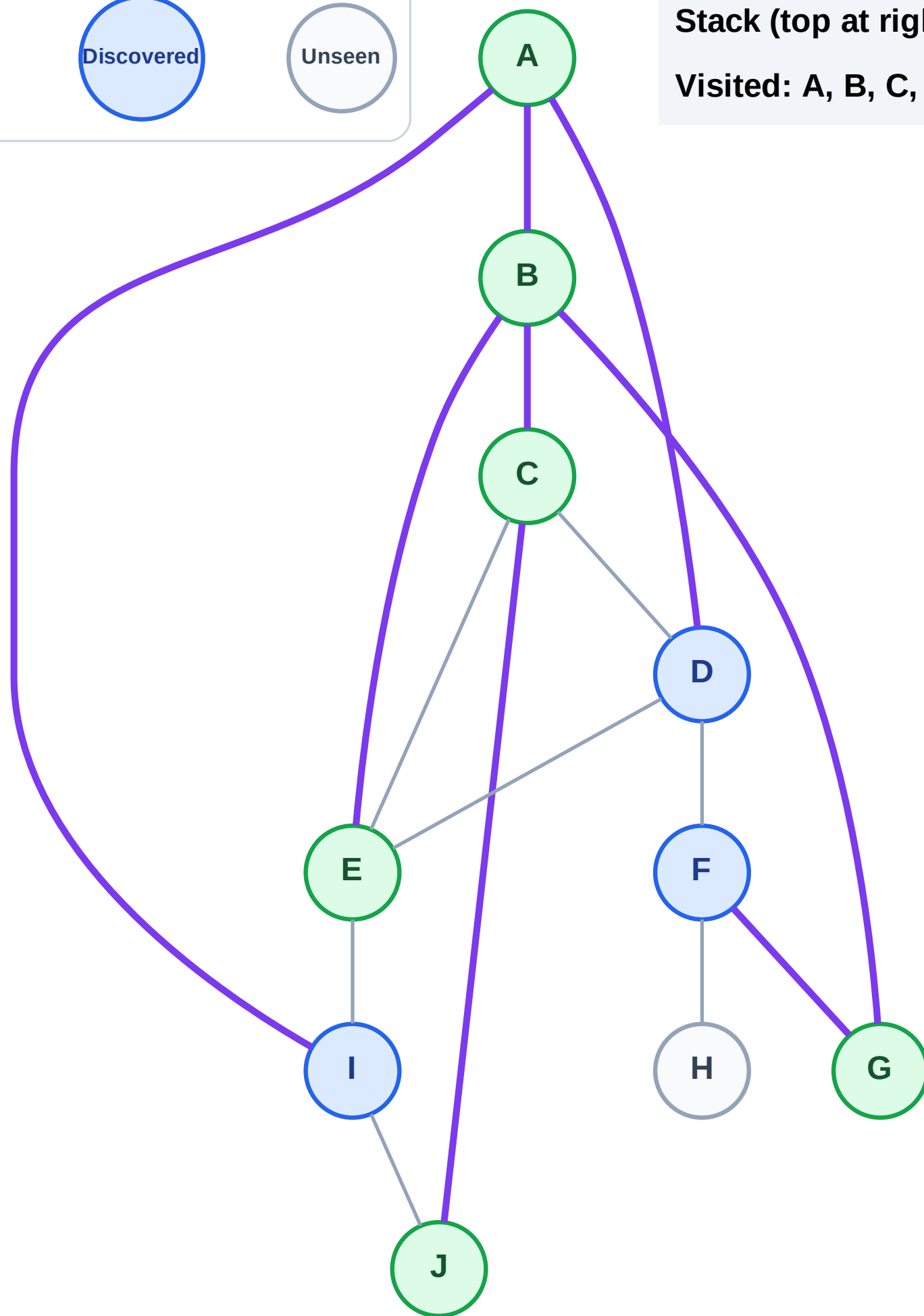
Step 13: Discover F from G

Legend



Stack (top at right): I | D | F

Visited: A, B, C, E, G, J



DFS Traversal

Step 14: Process node F

Legend

Complete

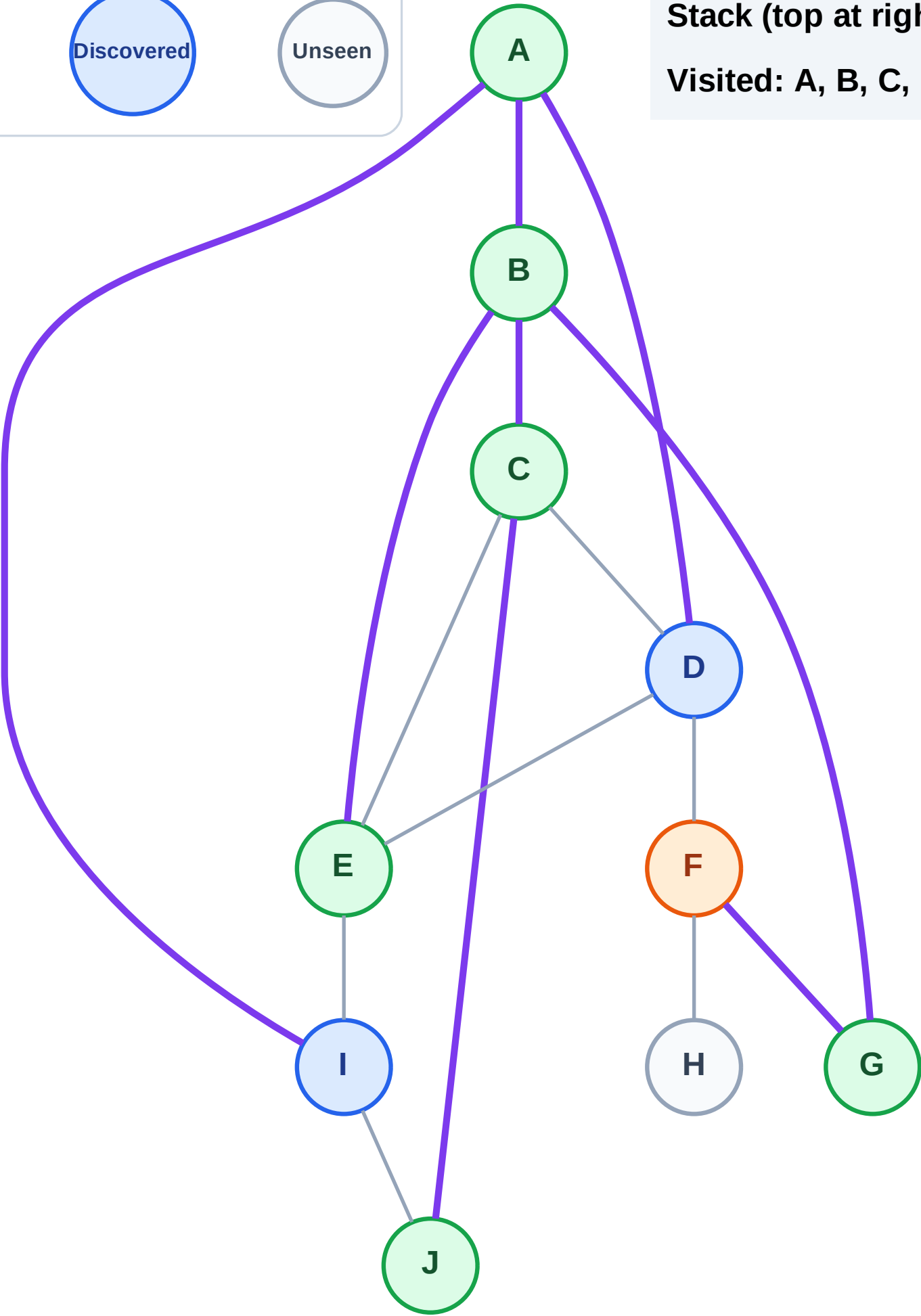
Processing

Discovered

Unseen

Stack (top at right): I | D

Visited: A, B, C, E, G, J



DFS Traversal

Step 15: Discover H from F

Legend

Complete

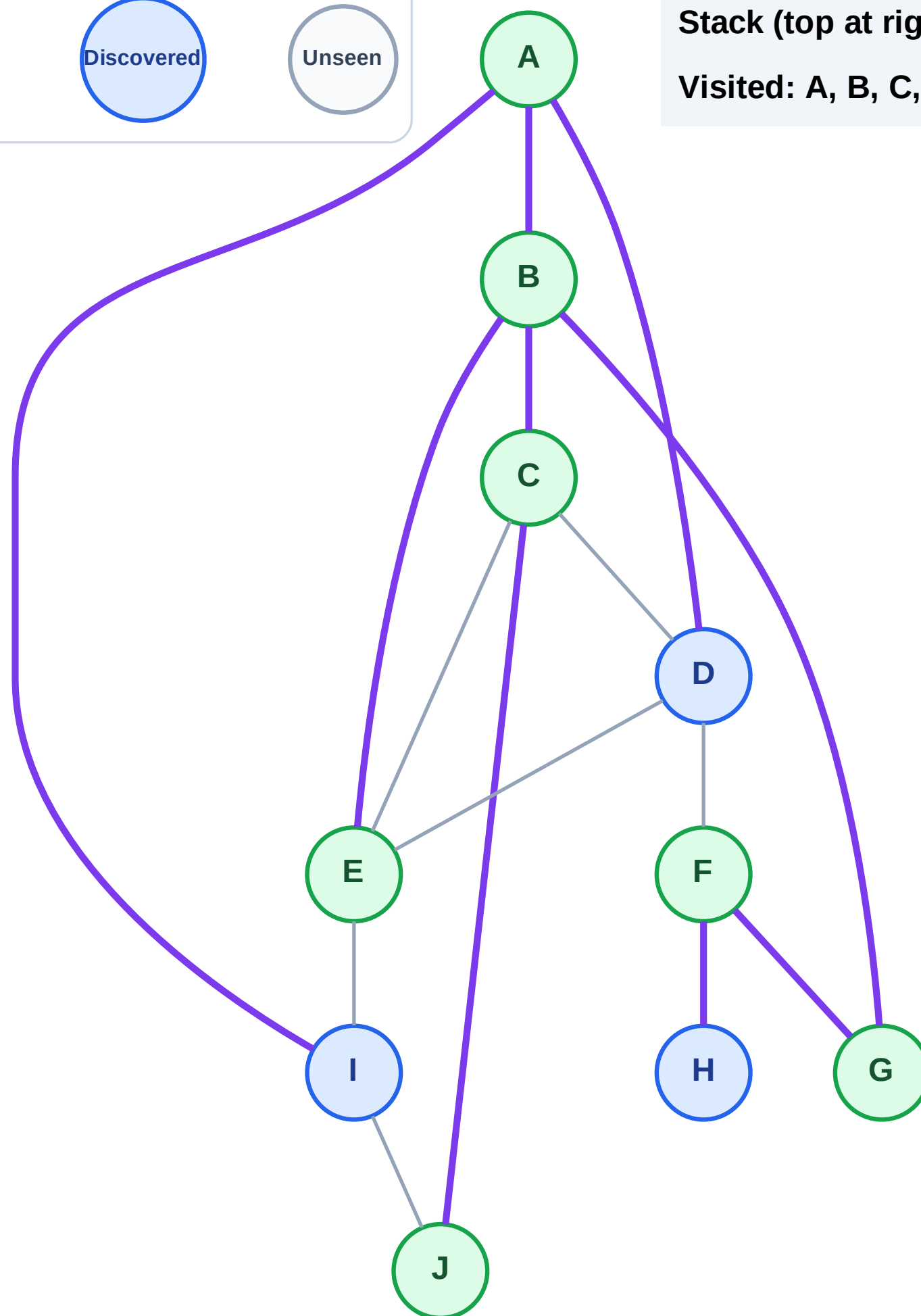
Processing

Discovered

Unseen

Stack (top at right): I | D | H

Visited: A, B, C, E, F, G, J



DFS Traversal

Step 16: Process node H

Legend

Complete

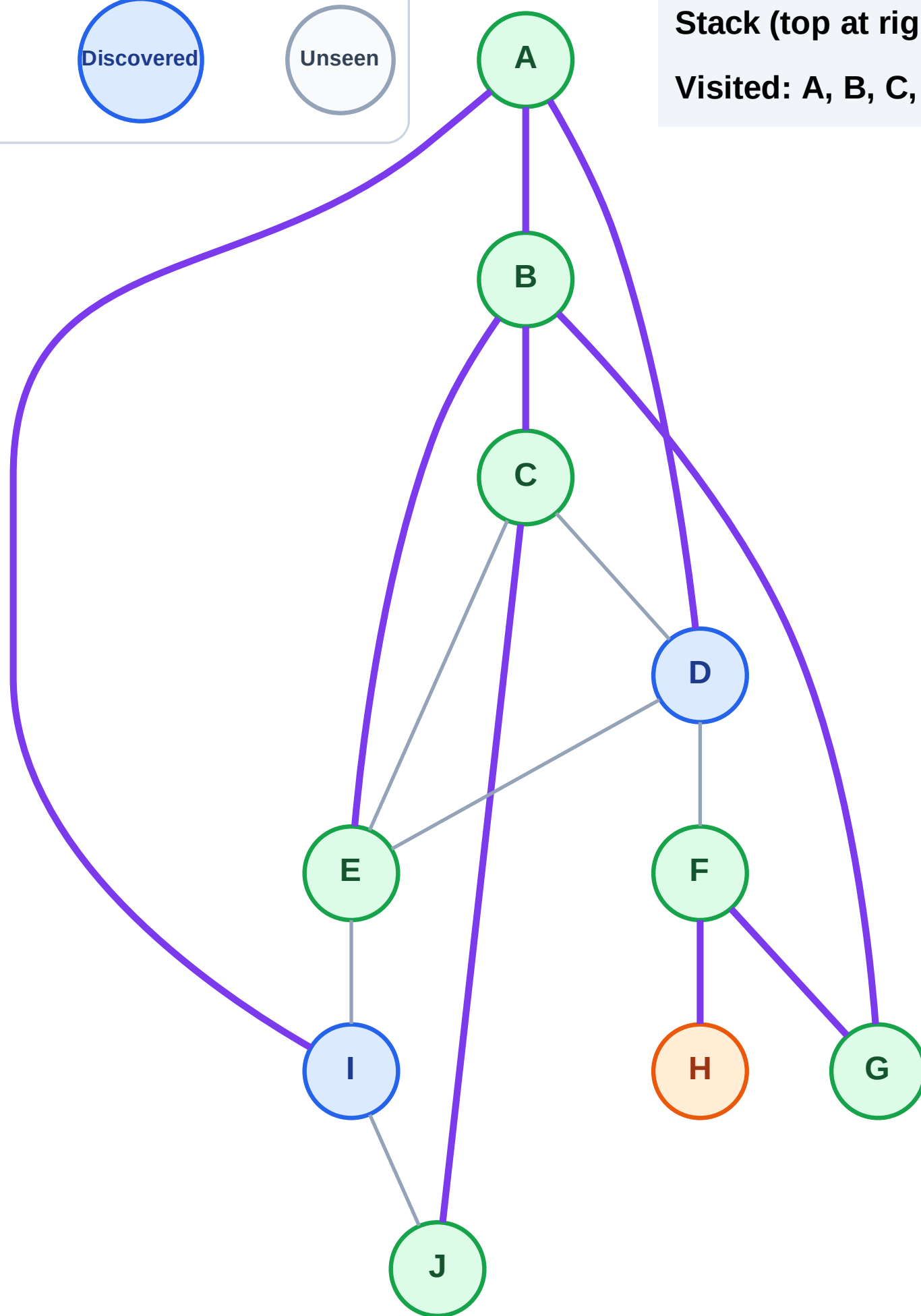
Processing

Discovered

Unseen

Stack (top at right): I | D

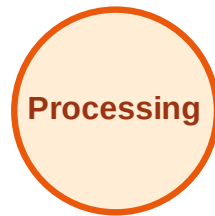
Visited: A, B, C, E, F, G, J



DFS Traversal

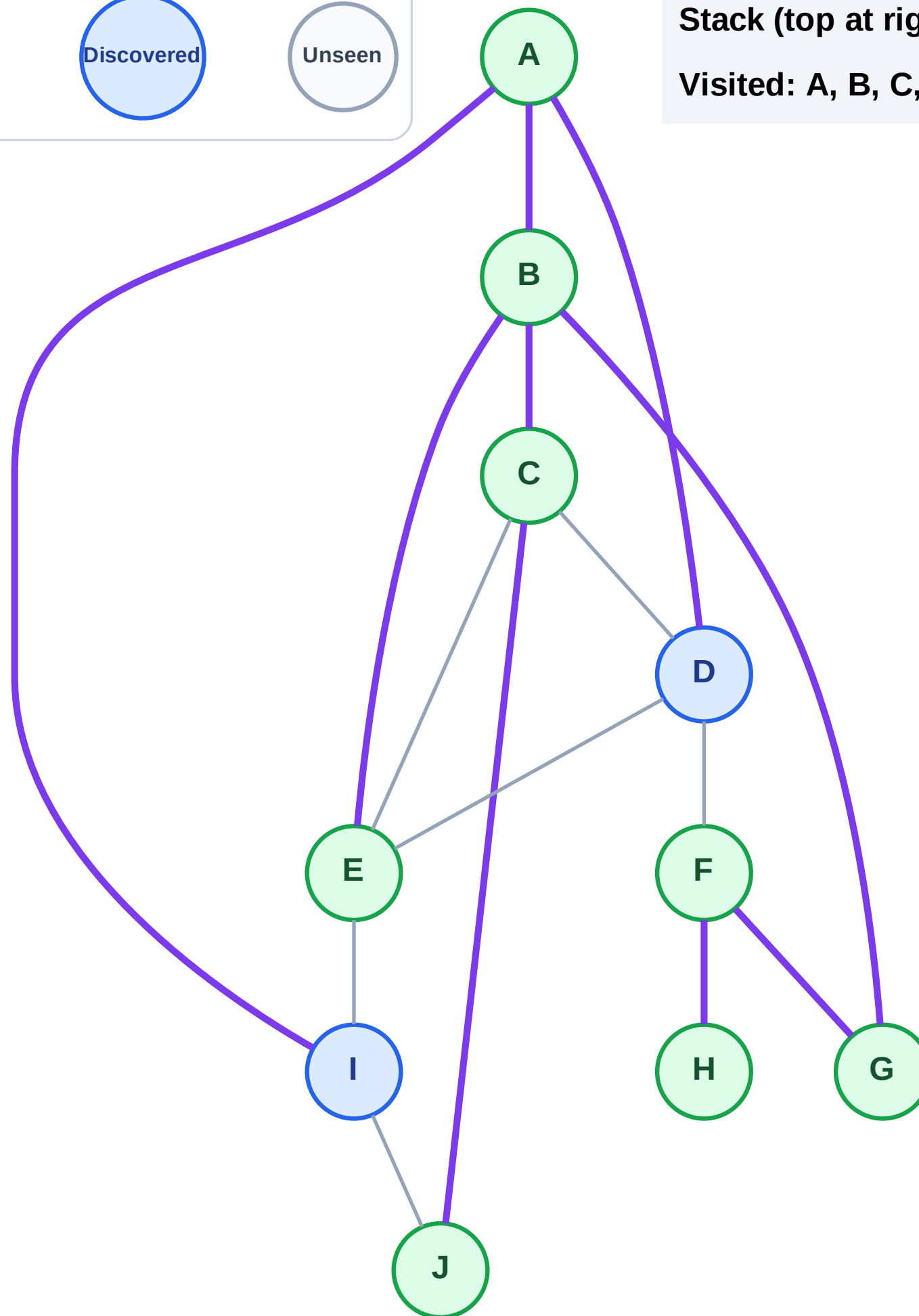
Step 17: No new neighbors from H

Legend



Stack (top at right): I | D

Visited: A, B, C, E, F, G, H, J



DFS Traversal

Step 18: Process node D

Legend

Complete

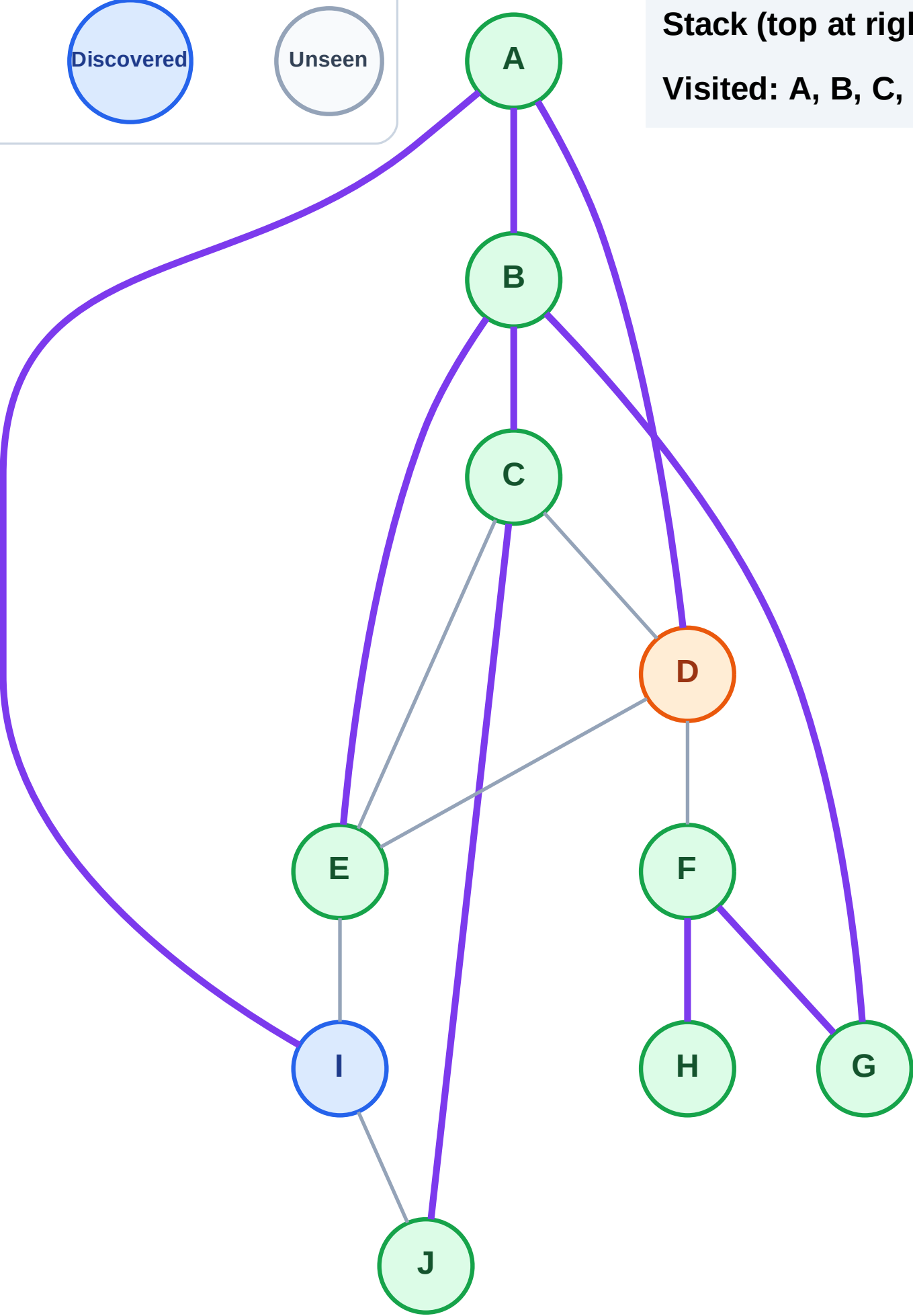
Processing

Discovered

Unseen

Stack (top at right): I

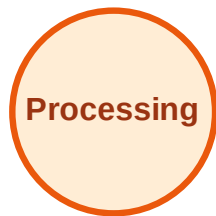
Visited: A, B, C, E, F, G, H, J



DFS Traversal

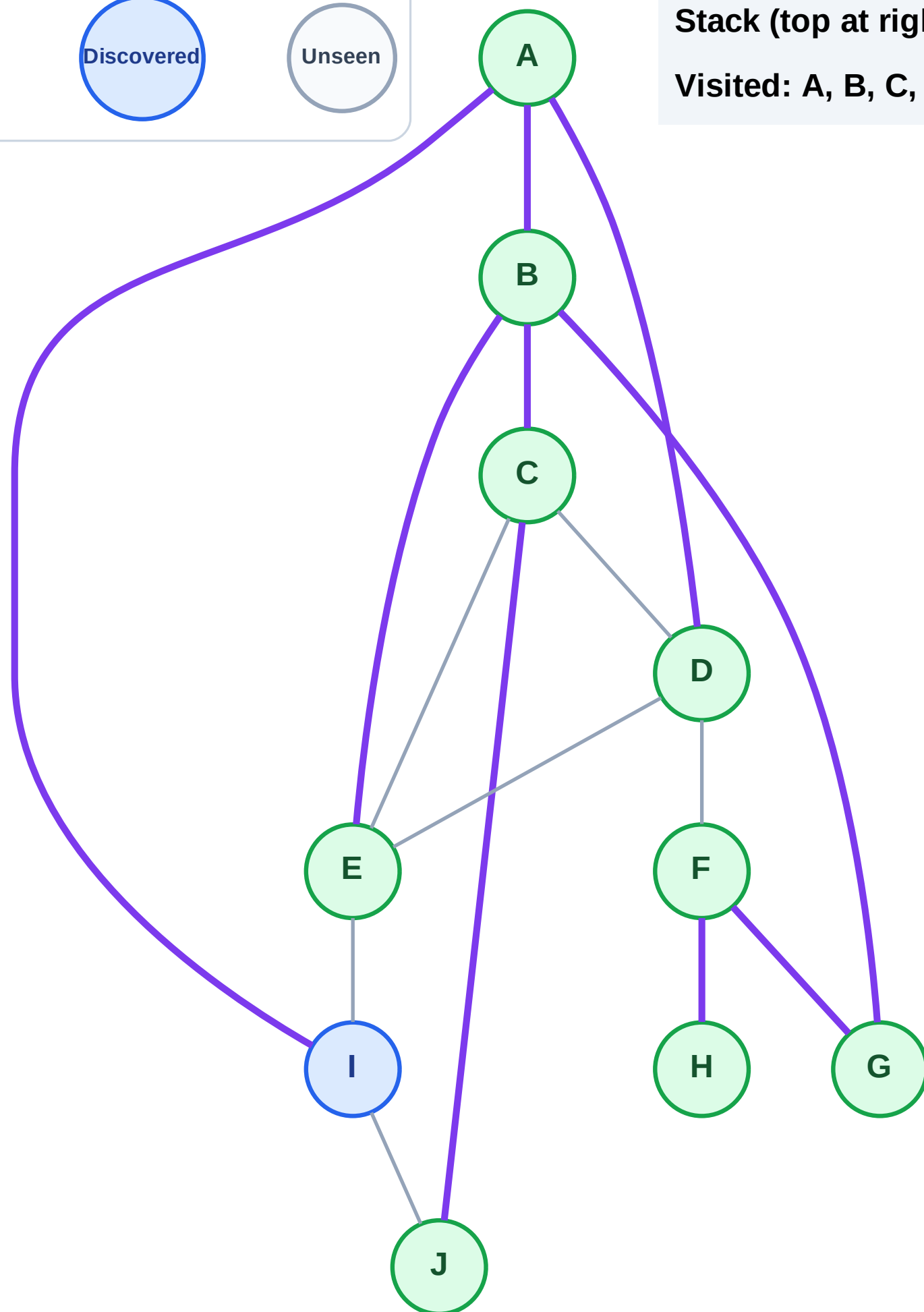
Step 19: No new neighbors from D

Legend



Stack (top at right): I

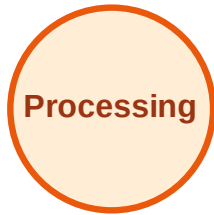
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

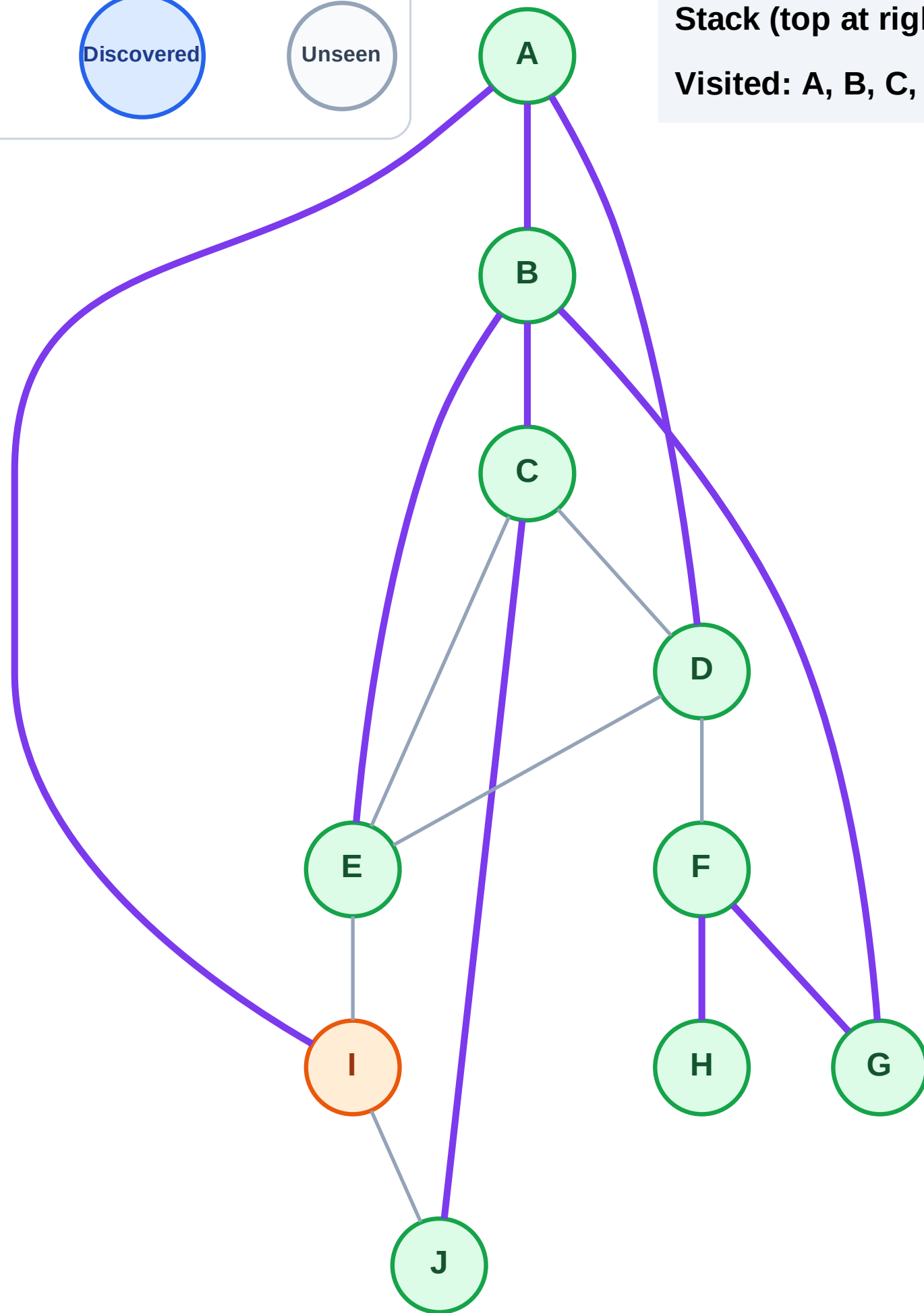
Step 20: Process node I

Legend



Stack (top at right): (empty)

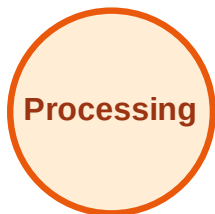
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

